



AFFORDABLE AUTOMATED FAN SYSTEM **FOR HUMIDITY REGULATION**

TECH PIONEERS

MT1940 – FUNDAMENTALS OF ENGINEERING DESIGN AND WORKSHOP
PRACTICE

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Summary

The main goal of our team's project is to find an automated cooling system that controls the unnecessary usage of electricity and also control the humidity. The reason to come across this idea is to avoid the electricity waste and discomfort caused by high heat.

Our project aims to address the challenge of finding an optimal cooling solution that is also cost-effective and highly efficient. We recognize that while inexpensive fans are readily available, they often fall short in terms of their effectiveness in providing adequate cooling. On the other hand, air conditioning machines offer remarkable cooling capabilities but come with a significant price tag, making them inaccessible for many individuals and organizations.

Through this project, we will focus on bridging this gap by developing an innovative cooling solution that combines affordability with exceptional performance. To avoid the unnecessary electricity consumption, we planned to use a sensor to indicate if someone is in a certain distance away from the cooling fan. And also if there is no movement around that distance for some time it is automatically turn off the fan. The time and the distance can be adjusted.

Instead of relying on expensive air conditioning, we use water to increase humidity naturally. With the help of the sensor, we can ensure that the system only runs when needed, preventing excessive moisture, and keeping the room comfortable.

Not only concerning our main problem but also our team has been concerned about following areas when designing this cooling system.

1. Sustainability
2. Cost
3. Materials
4. Ease of Design
5. Benefit
6. Safety in design

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1.INTRODUCTION

Air temperature variations in Sri Lanka exhibit distinct regional differences. In the lowlands, which encompass altitudes up to 100 to 150 meters, the mean annual temperature ranges from 26.5 °C to 28.5 °C, with an average of 27.5 °C. These days, the temperature even goes as high as 32 °C. But it is generally recommended to maintain temperatures between 23–25.5 °C (73–78 °F) for optimal comfort.

So, we searched about current and alternative cooling systems and future models. Air conditioning systems, fans, Evaporative Cooling System, Geothermal cooling system, Earth Air tunnel system, Adiabatic cooling system, Microclimate cooling system are some of them.

This helped us understand the current situation and challenges of cooling, to identify the needs and preferences of our target customers or users, to compare and evaluate the advantages and disadvantages of different cooling systems and to design and innovate a new cooling system that meets the criteria and expectations of our project.

An air conditioning system will work for up to 10 to 20 years. The lifespan of an air conditioning system can be impacted by frequency of use, maintenance, climate, installation quality and quality of the unit etc. A fan will work from 5 to 15 years on average. The lifespan of a fan can be impacted by frequency of use, brand and model of the fan, environmental factors, maintenance, power source and quality of the unit etc.

Prices of cooling systems range like this:

- Air conditioning systems: price range-Rs.160 000 - 500 000
- Fans: price range-Rs.7000- 50000
- Evaporative Cooling System (Portable): price range-Rs.20000 - 80000

For cooling systems to operate well and endure the intended amount of time, routine maintenance is necessary. The following actions can be taken to maintain a cooling system:

- Inspect and clean the air filters: Clogged air filters impede airflow, decreasing the effectiveness of the system. Every month or as needed, clean them or replace them.
- Inspect the condenser coils: Clogged coils can result in a 30% increase in energy usage. At least once a year, ideally just before the cooling season, clean them.
- Inspect the refrigerant level; low levels may be a sign of a leak. The system should be examined and recharged as needed by a competent technician.
- Inspect and clean the evaporator coils since dirty coils can impede airflow, extend the time the system runs, and lower efficiency. At least once a year, clean them.

- Check the ductwork for leaks, which can waste energy and lower system effectiveness. Use metal tape or mastic to seal them.
- Lubricate moving parts: To minimize wear and friction, moving parts like the fan motor and compressor require routine lubrication.
- Verify your thermostat: Verify the thermostat's calibration and functionality.
- Arrange for routine professional maintenance: At least once a year, a trained technician should inspect and maintain the system.
- You can assist guarantee that your cooling system is running effectively and extend its longevity by following these instructions.

The size of the room or hallway must be considered when choosing an air conditioner for that space. An air conditioner can typically chill 1 cubic meter of living space with 80 BTU (BTU stands for British Thermal Unit, a measurement of how much energy is used by an air conditioner to remove heat from indoor air).

While utilizing an air conditioner over time, some major issues could develop. Therefore, it needs to be fixed.

1. The absence of cooling or insufficient cooling in the room is the initial justification for repairing the a/c. There may be several causes for that issue.
 - As a result of dust and fiber buildup.
 - Freezing ice in the interior unit because of a drop in refrigerant gas pressure or a drop in compressor pressure.
 - Run errors when the condenser first starts.

The following procedures must be used to eliminate dust to prevent these issues.

1. Chemical service: The procedure of using chemicals to clean an air conditioner's interior parts, which also aids in removing collected debris from the unit. It might require chemical cleaning for a while, perhaps once every year or every two years.
2. Regular maintenance entails using brushes or a vacuum to clean the interior and outside of the appliance.

- The gas pressure needs to be checked, as well as any potential gas leaks.
- Verify your compressor. It must be replaced if it is worn out.

2. The other main problem is cooling stopped shortly after turning on the A/C. The following factors cause this to happen.

- Indoor unit fan blocked.
- Overload of the compressor in the outdoor unit due to lack of compressor oil.

To avoid the above problem,

- Check whether there is any difficulty running the fan and fix it.
- Refill the same compressor oil or replace it with a new compressor.

3. The other problem is electricity problems.

There can be some overloading. In this kind of situation, the whole electricity circuits in the both indoor and outdoor units in the air conditioner has to be checked.

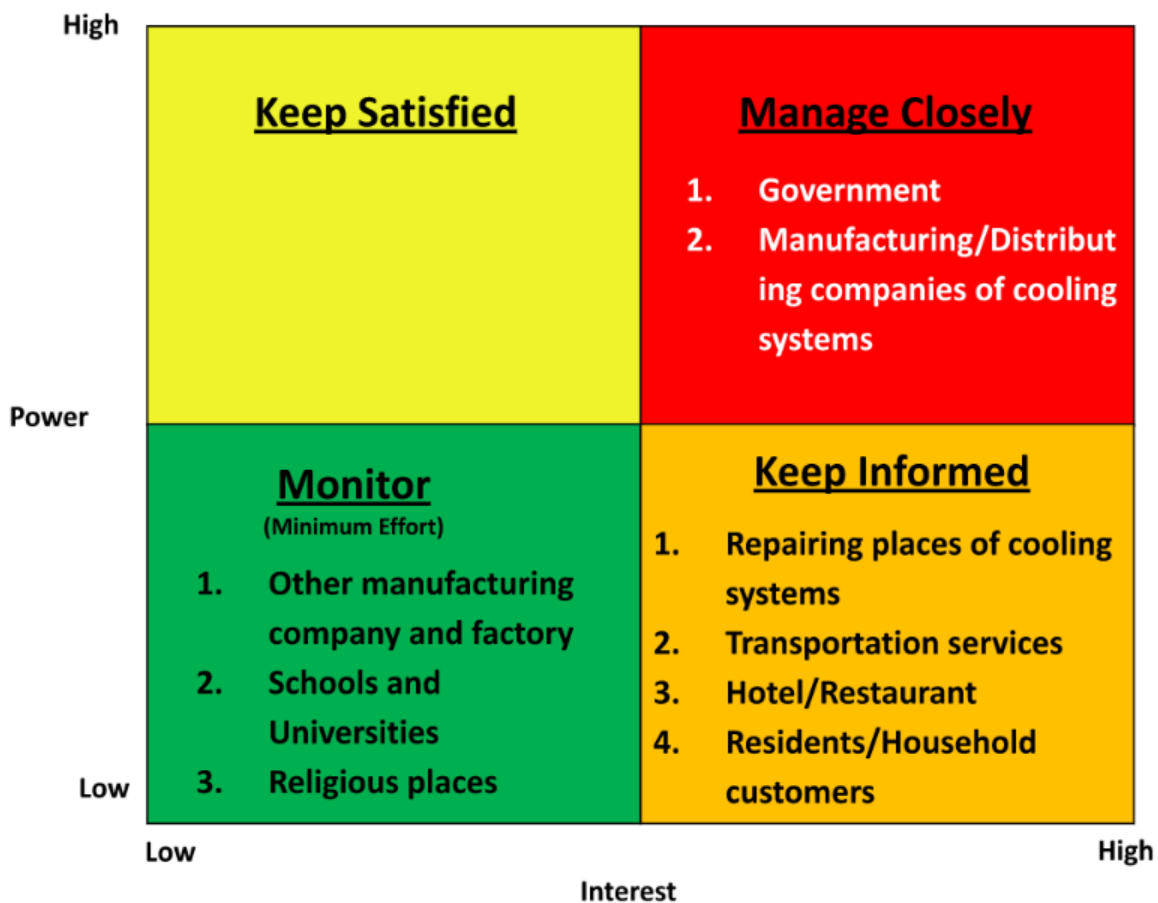
Some main problems that were identified in store bought cooling fans that already use the same method is that they were hard to clean and there was a risk of it being a mosquito breeding place. Also, the layer that is used to increase the humidity of the air cannot be removed, so it will get dirty and contain dust. So, it was planned to solve these problems and improve the design with an automated switch for the final design.

2.STAKEHOLDER ANALYSIS

2.1 Stakeholders

1. Government
2. Manufacturing/Distributing companies of cooling systems
3. Repairing places of cooling systems
4. Transportation services
5. Hotel/Restaurant
6. Residents/Household customers
7. Other manufacturing company and factory
8. Schools and Universities
9. religious places

2.2 Power – Interest Grid



2.3. Stakeholder Analysis

Stakeholder (Name/Images)	Position on Project (Explain how/why the stakeholder is connected with the problem)	Impact on Stakeholder (Explain how/why the stakeholder is impacted by the problem)
2.3.1. Government 	Key stakeholder with the authority to establish regulations, policies, and initiatives: The government holds a critical position in shaping the regulatory framework and implementing policies related to energy efficiency, environmental sustainability, and public welfare. It has the power to create an enabling environment for the development and adoption of innovative technologies. Responsible for energy efficiency, environmental sustainability, and public welfare: The government has the responsibility to prioritize energy efficiency measures that reduce energy consumption and promote sustainable practices. It aims to protect the environment and ensure the well-being of its citizens by implementing policies that address societal challenges such as climate change and affordability of essential services.	<u>If solution was found:</u> Reduced Energy Consumption: The implementation of the affordable automated fan system would contribute to a significant reduction in energy consumption. These systems are designed to operate efficiently, consuming less energy compared to traditional cooling systems. The government would achieve its energy conservation goals and support its initiatives to combat climate change by promoting energy-efficient technologies. Environmental Sustainability: The affordable automated fan system aligns with the government's sustainability objectives. It would lead to a decrease in carbon emissions and a smaller ecological footprint. By encouraging the adoption of this system, the government can demonstrate its commitment to mitigating the environmental impacts associated with conventional cooling systems.

	Aims to improve living conditions and promote cost-effective solutions: The government strives to enhance the overall living conditions of its citizens. It seeks to promote cost-effective solutions that address the needs of the public, including affordable technologies for humidity regulation. By identifying and supporting such solutions, the government can contribute to improving the quality of life for its citizens. Governing body with the power to drive adoption of innovative technologies: As a governing body, the government holds the power to drive the adoption of innovative technologies. By establishing favorable policies, providing incentives, and creating a supportive ecosystem, the government can encourage the adoption of the affordable automated fan system for humidity regulation. This can lead to widespread implementation and positive outcomes for both the public and the environment.	Cost-Effective Solution for the Public: The availability of an affordable automated fan system would provide cost-saving benefits for the public. It would allow households, educational institutions, religious places, and other establishments to regulate humidity levels efficiently without incurring exorbitant expenses. The government's efforts to enhance public welfare and affordability would be supported through the implementation of this system. <u>If solution was not found:</u> If our cooling device is not successful, it may have a negative impact on the government. Reputation and reliability damage: If the government does not take adequate measures to make cooling devices more affordable and accessible to the common public, it will damage their reputation and reliability among the citizens. People will perceive the government as uncaring and out of touch with the day to day struggles of the masses. This can seriously undermine trust in
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		the government and their policies. Hindered Energy Efficiency Goals: The government's goals related to energy efficiency and sustainability would face challenges without the adoption of the affordable automated fan system. The inability to provide a cost-effective alternative could impede progress in reducing energy consumption and carbon emissions. Delayed Environmental Benefits: Without the implementation of this system, the government's efforts to promote environmentally friendly practices and mitigate climate change might be delayed. The lack of a viable solution could result in the continued reliance on conventional cooling systems that have higher energy requirements and contribute to environmental degradation. Limited Cost Savings for the Public: The absence of an affordable automated fan system would limit cost-saving opportunities for households, institutions, and businesses. The public would continue to rely on
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		less efficient cooling systems, leading to higher energy bills and increased expenses. This could hinder the government's objective to improve affordability and the standard of living for its citizens.
2.3.2.Manufacturing/ Distributing companies 	Prospects for new market opportunities: Manufacturing companies recognize the potential for new market opportunities in the development of an affordable automated fan system for humidity regulation. If they can actively participate in the design and manufacturing processes, it may open doors to increased revenue streams and market expansion. Potential threat to existing products: Manufacturing companies are aware that the introduction of an affordable automated fan system for humidity regulation could pose a potential threat to their existing products. If the new system gains popularity, it may lead to a decline in sales and revenue for traditional humidity regulation devices. Need for adaptation to new technology:	<u>If a solution is not found:</u> Loss of revenue and profitability: Without a solution, manufacturing companies could face declining sales and revenue as customers seek alternative, more affordable humidity regulation options. Limited market competitiveness: Failure to adapt to new technologies and automation systems may result in reduced competitiveness compared to companies that successfully implement affordable automated fan systems. Potential job losses: In the absence of a solution, manufacturing companies may face the risk of downsizing or layoffs due to reduced demand for traditional humidity regulation devices. <u>If a solution is found:</u>

	Manufacturing companies must be willing to adapt to new technologies and automation systems to participate effectively in the development and production of an affordable automated fan system. This may require investments in advanced manufacturing equipment, software, and employee training.	New revenue opportunities: Manufacturing companies involved in the production of affordable automated fan systems can benefit from increased revenue and market share by capitalizing on the demand for cost-effective humidity regulation solutions. Technological advancement and expertise: Participation in the development of an affordable automated fan system enables manufacturing companies to acquire new technological capabilities and expertise, positioning them as leaders in the industry. Job creation and workforce development: Successful implementation of the new system may lead to job creation as manufacturing companies expand their workforce to meet the increased demand for the affordable automated fan system. Enhanced competitiveness: Manufacturing companies actively involved in the development and manufacturing of the
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		affordable automated fan system can gain a competitive edge by offering a cost-effective solution that meets customer demands.
2.3.3.Repairing places 	Potential for New Service Opportunities: Repairing places recognize the potential for new service opportunities in relation to the affordable automated fan system for humidity regulation. The introduction of a new technology may lead to increased demand for installation, maintenance, and repair services specifically related to the automated fan system. Technological Adaptation and Training: Repairing places may need to adapt to new technologies and acquire the necessary expertise to effectively handle the installation, maintenance, and repair of the affordable automated fan system. This may require investing in training programs and updating tools and equipment. Impact on Existing Services: The development of the affordable automated fan system may impact the demand for repairing	<u>If solution is not found</u> Limited Service Opportunities: Without a solution, repairing places may face limited service opportunities related to the affordable automated fan system. This could potentially restrict their business growth and revenue potential. Need for Diversification: Repairing places may need to diversify their services to include repairs and maintenance of alternative humidity regulation systems to compensate for the potential decline in demand for repairs on traditional devices. 2.2 If solution is found New Service Opportunities: Repairing places that are knowledgeable and skilled in the installation, maintenance, and repair of the affordable automated fan system can benefit from increased service opportunities. This can contribute to their business growth and revenue generation.

	services related to traditional humidity regulation devices. Repairing places may need to assess the potential decline in demand for repairs on older systems and adapt their services accordingly.	Technological Expertise: Successful participation in the implementation of the affordable automated fan system can enhance repairing places' technological expertise and capabilities. This positions them as knowledgeable service providers and may attract more customers seeking professional assistance. Need for Continuous Training: As new technologies and advancements emerge, repairing places will need to invest in continuous training to stay updated and competent in handling the affordable automated fan system. This ensures they can provide quality services and maintain customer satisfaction.
2.3.4. Transportation services 	Potential Impact on Distribution and Delivery: Transportation services recognize the potential impact the development of an affordable automated fan system for humidity regulation may have on their distribution and delivery operations. Depending on the size, weight, and handling requirements of the new	<u>If a solution is not found</u> Limited Growth Opportunities: Without a solution, transportation services may face limited growth opportunities specifically related to the delivery of the affordable automated fan system. This could restrict their potential to expand their services in this particular market segment.

	system, transportation services may need to adapt their logistics to accommodate these changes. Efficiency and Cost Considerations: Transportation services are interested in the efficiency and cost implications of delivering the affordable automated fan system. If the system is lightweight, compact, and easy to handle, transportation services may benefit from reduced shipping costs and streamlined delivery processes. Potential for Increased Demand: Transportation services may anticipate increased demand for their services as the affordable automated fan system gains popularity and more units need to be transported to various locations. This could present opportunities for business growth and increased revenue.	Reduced Efficiency and Cost Savings: If the affordable automated fan system is not designed with transportation efficiency in mind, transportation services may face challenges related to handling, storage, and transportation costs. This could impact their operational efficiency and reduce potential cost savings. <u>If a solution is found</u> Increased Delivery Demand: Transportation services can expect increased demand for their delivery services as the affordable automated fan system gains popularity. This can lead to expanded business opportunities and revenue growth. Streamlined Logistics and Cost Savings: If the affordable automated fan system is designed with transportation efficiency in mind, transportation services can benefit from streamlined logistics, reduced handling efforts, and potential cost savings in terms of packaging, storage, and transportation.
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		Collaboration Opportunities: Transportation services may have opportunities for collaboration with manufacturers and distributors of the affordable automated fan system to optimize delivery processes, implement just-in-time strategies, and enhance overall supply chain efficiency.
2.3.5.Hotel/Restaurant Owners 	Need for Efficient Humidity Control: Hotel and restaurant owners recognize the importance of humidity regulation in maintaining guest comfort and food quality. They are interested in adopting cost-effective solutions like the affordable automated fan system to ensure optimal humidity levels within their establishments. Cost Reduction and Operational Efficiency: Hotel and restaurant owners are interested in solutions that can help reduce operational costs, including energy expenses. The affordable automated fan system has the potential to provide energy-efficient humidity regulation, leading to cost savings in the long term. Guest Satisfaction and Comfort:	<u>If a solution is not found</u> Increased Operating Costs: Without a solution, hotel and restaurant owners may continue to rely on traditional humidity regulation devices that consume more energy and result in higher operating costs. This can impact their profitability and financial performance. Potential Comfort and Quality Issues: Traditional humidity regulation devices may not provide precise and efficient control over humidity levels, leading to potential discomfort for guests and compromising the quality of food and beverages served. This can negatively impact the reputation of hotels and restaurants. <u>If a solution is found</u>

	Providing a comfortable and pleasant environment for guests is a priority for hotel and restaurant owners. The implementation of an effective humidity regulation system, such as the affordable automated fan system, can contribute to guest satisfaction and positive experiences.	Cost Savings and Operational Efficiency: The affordable automated fan system can contribute to cost savings through improved energy efficiency. Hotel and restaurant owners can benefit from reduced energy expenses and enhanced operational efficiency, resulting in increased profitability. Enhanced Guest Experience: By maintaining optimal humidity levels, the affordable automated fan system can contribute to a comfortable and pleasant environment for guests. This can lead to improved guest satisfaction, positive reviews, and increased customer loyalty. Environmental Sustainability: The implementation of the affordable automated fan system aligns with the growing emphasis on environmental sustainability. Hotel and restaurant owners can promote their commitment to sustainability and attract environmentally conscious customers.
2.3.6.Hotel/Restaurant Customers	Affordability and Value for Money:	<u>If solution is not found</u> Potential Discomfort:



Customers are interested in affordable options that provide value for money. They appreciate cost-effective solutions that deliver efficient humidity regulation without significantly impacting their expenses.

Environmental Consciousness:

Increasingly, customers are concerned about environmental sustainability and appreciate businesses that prioritize eco-friendly practices. They may value establishments that implement energy-efficient solutions like the affordable automated fan system.

Traditional humidity regulation systems may not always provide precise control over humidity levels, resulting in potential discomfort for customers. This can impact their overall experience and satisfaction.

Limited Cost Savings:

Without a cost-effective solution, customers may not benefit from potential cost savings in the form of lower energy expenses, which could indirectly affect their overall expenses and value for money.

If solution is found

Enhanced Comfort and Satisfaction:

The implementation of the affordable automated fan system can contribute to a more comfortable environment with optimal humidity levels. This enhances the overall experience and satisfaction of hotel and restaurant customers.

Potential Affordability and Value for Money:

The affordable automated fan system can result in cost savings for businesses, which may translate into more affordable prices or enhanced value for money for customers. They can

		enjoy efficient humidity regulation without significant price increases. Support for Environmental Sustainability: Customers who prioritize environmental sustainability appreciate establishments that implement energy-efficient solutions like the affordable automated fan system. This aligns with their values and provides an added incentive to choose environmentally conscious hotels and restaurants.
2.3.7.Residents/Household customers 	Comfort, Health, and Well-being: Customers prioritize comfort, health, and well-being within their homes. They value solutions that provide optimal humidity regulation to create a comfortable living environment, prevent mold and mildew growth, and minimize respiratory issues and allergies associated with poor indoor air quality. Energy Efficiency and Cost Savings: Customers are increasingly concerned about energy consumption and its impact on the environment and their expenses. They seek energy-efficient solutions that not only provide effective humidity	<u>If a solution is not found,</u> Discomfort and Health Concerns: Without an efficient humidity regulation system, residents may experience discomfort due to excessive moisture or dryness in their homes. High humidity can lead to a stuffy and unpleasant environment, while low humidity can cause dry skin, throat irritation, and respiratory issues. Increased Energy Consumption and Expenses: Traditional humidity regulation devices may consume more energy,

	regulation but also contribute to reduced energy consumption, resulting in cost savings and a smaller carbon footprint. Convenience and User-Friendliness: Customers prefer solutions that are convenient and user-friendly. They appreciate automated fan systems that offer ease of use, intuitive controls, and minimal maintenance requirements. A hassle-free experience enhances their satisfaction and encourages adoption of the system.	resulting in increased energy consumption and higher utility bills for residents. This can strain their household budgets and limit their ability to manage expenses effectively. Environmental Impact: Higher energy consumption associated with traditional humidity regulation devices contributes to increased greenhouse gas emissions and environmental degradation. Residents may feel a sense of responsibility to reduce their carbon footprint but face limitations without an energy-efficient alternative. <u>If a solution is found,</u> Improved Comfort and Indoor Air Quality: The implementation of the affordable automated fan system can lead to improved comfort by maintaining optimal humidity levels in residents' homes. This creates a pleasant living environment and promotes better indoor air quality, reducing the risk of mold, mildew, and respiratory issues. Energy Savings and Cost Efficiency:
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		The affordable automated fan system, designed for energy efficiency, can help residents reduce their energy consumption. This not only lowers their utility bills but also contributes to overall cost efficiency, allowing them to allocate their resources more effectively. Environmental Responsibility: Residents who prioritize environmental sustainability appreciate the energy-efficient design of the affordable automated fan system. By choosing this solution, they actively participate in reducing their carbon footprint and promoting eco-friendly practices within their households. Convenience and Ease of Use: Automated fan systems offer convenience and ease of use for residents. With intuitive controls and automated features, residents can effortlessly regulate humidity levels, ensuring optimal conditions without the need for manual adjustments or extensive technical knowledge.
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2.3.8. Other manufacturing company and factory



Need for Effective Cooling Solutions:

Manufacturing companies and factories prioritize the need for effective cooling solutions to maintain optimal working conditions for employees and protect sensitive equipment. They are interested in adopting affordable automated fan systems that can provide efficient cooling while minimizing costs.

Energy Efficiency and Cost Reduction:

Manufacturing companies and factories are increasingly concerned about energy consumption and its impact on operational expenses. They seek energy-efficient cooling solutions that can help reduce energy consumption, lower utility bills, and improve overall cost-effectiveness.

Customizability and Adaptability:

Manufacturing companies and factories often require cooling systems that can be customized and adapted to their specific requirements. They value fan products that offer flexibility in terms of installation, placement, and control to ensure seamless integration

If solution is not found

Higher Energy Costs:

Without an energy-efficient cooling solution, manufacturing companies and factories may experience increased energy costs due to the use of traditional cooling methods. This can negatively impact their operational expenses and overall profitability.

Potential Equipment Damage and Downtime:

Insufficient or inefficient cooling systems can lead to overheating of equipment and machinery, resulting in potential damage and costly downtime for repairs or replacements. This can disrupt production schedules and impact productivity.

Limited Flexibility and Adaptability:

Lack of customizable cooling solutions can limit manufacturing companies and factories in terms of effectively integrating cooling systems into their existing infrastructure. This can lead to inefficiencies and potential disruptions in the production process.

	into their production processes.	<u>If solution is not found</u> Energy Savings and Cost Reduction: The implementation of an energy-efficient automated fan system can result in significant energy savings for manufacturing companies and factories. This can lower operational costs, improve cost-effectiveness, and enhance overall profitability. Equipment Protection and Increased Productivity: Efficient cooling systems contribute to equipment protection, minimizing the risk of overheating and reducing the likelihood of costly equipment downtime. This helps maintain smooth operations, increase productivity, and optimize production output. Customization and Adaptability: An affordable automated fan system that offers customization and adaptability allows manufacturing companies and factories to tailor the cooling solution to their specific needs. This flexibility enables seamless integration into their existing infrastructure and production processes.
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2.3.9.Schools and Universities



Temperature Control for Learning Environments:

Schools and universities prioritize maintaining optimal temperature conditions within classrooms, lecture halls, and other learning spaces. They are interested in adopting affordable automated fan systems that can provide effective cooling to create comfortable and conducive learning environments.

Energy Efficiency and Operational Costs:

Educational institutions have a strong focus on energy efficiency and cost reduction. They seek cooling solutions that are energy-efficient to minimize their environmental impact and reduce operational expenses associated with cooling systems.

Noise and Disruption Considerations:

Schools and universities require cooling solutions that do not generate excessive noise or disrupt teaching and learning activities. They value fan products that offer quiet operation to ensure a peaceful environment for students and faculty.

If solution is not found:

Reduced Learning Environment Quality:

Insufficient cooling systems can lead to uncomfortable and distracting learning environments for students and faculty. This may affect concentration, engagement, and overall learning outcomes.

Higher Operational Costs:

Traditional cooling systems may result in higher energy consumption and increased operational costs for schools and universities. Limited budget resources may be diverted towards cooling expenses, reducing the availability of funds for educational programs and initiatives.

Maintenance and Repair Challenges:

Outdated or inefficient cooling systems can require frequent maintenance and repairs, leading to disruptions in regular school activities and potential additional costs for repairs or replacements.

If a solution is found:

Enhanced Learning Environment:

		The implementation of an effective automated fan system can significantly improve the learning environment, ensuring comfortable temperatures that support concentration, focus, and overall academic performance. Energy Savings and Cost Reduction: Energy-efficient cooling solutions contribute to reduced operational costs for schools and universities. Lower energy consumption can free up budget resources for other educational purposes and initiatives. Minimal Disruption and Noise Levels: An affordable automated fan system that operates quietly minimizes noise disruptions during teaching and learning activities. This creates a peaceful environment conducive to effective communication and concentration. Sustainability and Environmental Responsibility: The adoption of energy-efficient cooling systems aligns with schools' and universities' commitment to sustainability and environmental responsibility. It
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		demonstrates their dedication to reducing carbon footprint and promoting eco-friendly practices.
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2.3.10. Religious places



Importance of Comfortable Worship Environment:

Religious places prioritize providing a comfortable worship environment for their congregations. They are interested in adopting affordable automated fan systems that can effectively cool the spaces and enhance the overall experience of worshippers.

Preservation of Sacred Structures and Artifacts:

Religious places often have valuable sacred structures, artifacts, and artworks that need protection from excessive heat and humidity. They seek cooling solutions that can maintain optimal temperature and humidity levels to preserve these important religious and cultural assets.

Energy Efficiency and Sustainable Practices:

Religious places have a growing concern for energy efficiency and sustainable practices. They are interested in cooling solutions that are energy-efficient, minimizing their environmental impact and supporting their commitment to responsible resource management.

If a solution is not found:

Discomfort for Congregations:

Insufficient cooling systems can lead to discomfort among congregations during religious ceremonies and gatherings. High temperatures or excessive humidity can hinder concentration and detract from the worship experience.

Risk to Sacred Structures and Artifacts:

Inadequate cooling systems may expose sacred structures, artifacts, and artworks to the risk of damage due to temperature fluctuations and excess moisture. This can result in the deterioration of cultural and religious heritage.

Increased Energy Consumption:

Traditional cooling methods may lead to higher energy consumption and increased operating costs for religious places. This can strain their financial resources and impact their ability to allocate funds to other essential areas.

		<u>If a solution is found:</u> Enhanced Worship Experience: The implementation of an effective automated fan system can create a comfortable worship environment for congregations, allowing them to focus on their spiritual practices and rituals without distraction from uncomfortable temperatures. Preservation of Sacred Structures and Artifacts: An affordable automated fan system designed to maintain optimal temperature and humidity levels can contribute to the preservation of sacred structures, artifacts, and artworks. This ensures their longevity and protects the cultural and religious heritage of the place of worship. Energy Efficiency and Sustainability: Energy-efficient cooling solutions align with the sustainability goals of religious places. By adopting such systems, they demonstrate their commitment to responsible energy consumption and contribute to environmental conservation efforts.
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3.Design Requirements

Design Criteria

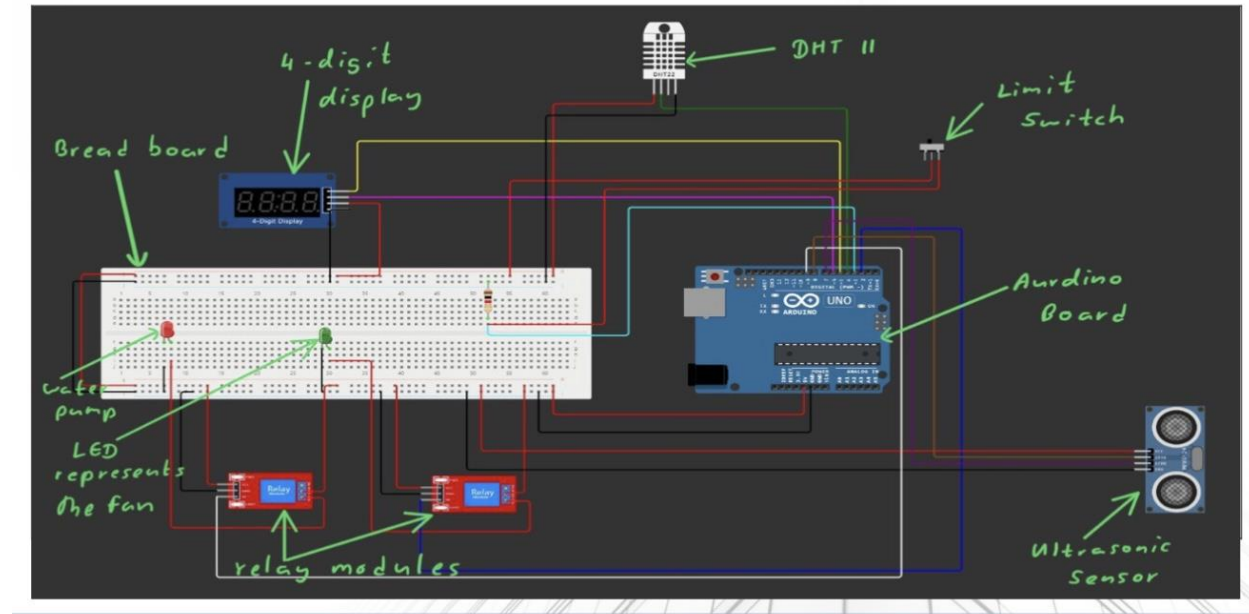
Factors to consider	Stakeholder's Wants/Needs	Design Requirements	
		Criteria	Constraints
3.1. Sustainability (10 marks)	Minimal resources 2 Reduce the environmental pollution 2 Minimal Waste 2 Use of waste products from the other processes 2 Ability to process the raw materials in an environmentally friendly way 2	Should reduce the environmental pollution Should use waste products from other processes Should use sustainable resources Should reduce wastage Should reduce the impact to the human health Should increase the productiveness of the users from various cultures and social status	• must abide by the regulations and norms of environmental policies. • It must be accessible to the whole public financially. • Must not contain any offensive language in its design, including insults, racist or religious symbols. • Insects, animals which are living nearby must be not affected by this process..
3.2. Cost (20 mark)	Comfortable price 10 Less expensive resources and maintenance cost 10	Should reduce the maintenance cost Should use the less expensive resources to maintain a bearable price	• Should not exceed the budget
3.3. Materials (20 marks)	Local materials 5 Ability to process the materials easily 5	Should use eco-friendly and less expensive materials Should be able to take another local	• Must use local and easy to find materials. • Must not use hazardous materials for people

	Inexpensive and easy to find 5 Minimal resources 2 Ability to use in most of the environmental conditions 3	material as substitute Should use materials which are suitable for most of the local environmental conditions Should use easily and safely processable materials	
3.4.Ease of Design (20 marks)	Easy to use and repair 5 Capability of use in every environment situation 2 Ability to transport easily and safely 3 Less space 5 Better temperature control 5	Should be capable of using in every environmental situation Should be easy to use and repair Should be easy and safe to transport. Should take minimal space	• Lids must be airtight and sealed. • The entire space in the container must be useful. • Insects should not enter to the box • Cover all the entrances.
3.5.Benefit (20 marks)	Energy Efficiency 10 High capacity 5 Multi-purpose 5	Should be energy efficient The capacity should be high	• Must increase the humidity of the air.
3.6.Safety in design (10 marks)	If fire is involved, confine and manage that safely in order to prevent burn injuries, house fires 2 Reduce the probability of blasts happening 2 Ability to use without exposure to toxic substances 2	Should ensure extra safety in design Should be able to safely use by everyone Should terminate the equipment in an emergency situation Should reduce the damage to the	• Must not produce carcinogenic substances • No dangerous elements, such as extremely flammable materials, may be used in the design.

	Special safety features for children (ex: Child lock) 4	equipment in an accident Should not produce dangerous substances like toxic gases	
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4.Design Solution

The solution we proposed is an electrically efficient, air cooling system. Unlike traditional air cooling systems, we; the Tech pioneers introduced special and some cool features for our cooling solution.



- Limit switch, Ultrasound sensor, DHR 11 sensors were used to develop the Design solution. Moreover relay modules were attached to Bread board and also an arduno board has been used



too. There are two LEDs attached to bread board that indicates the functionality of the cooling fan and the water pump.

- The cooling system is automated to identify an user and turn on and off automatically. 90cm distance was used to identify whether the user is there or not. This helps to save the electricity in an efficient manner.
- When speaking about Automated features, a sudden problem may pop up. That's if the Cooler is Automated and the user moves away for a quick task, that may occur a problem too. So the design has set an 10s timer that to turn off the cooler. We can change the timing period as same as the distance as user requires.
- The conceptual design solution also provided a digital display that displays the humidity level of the room. That can be use as an indicator for the user that to note his/her comfortable humidity level or to turn on/off the cooler manually.
- For a proper air flow, air ventilation holes were made at the side and also back of the cooler has been covered with a net.

5. Prototype specification

5.1. The process of making structure of the cooling system

Materials used:

Outer structure

- Hardboard
- Gum tape
- Gule

Inner structure

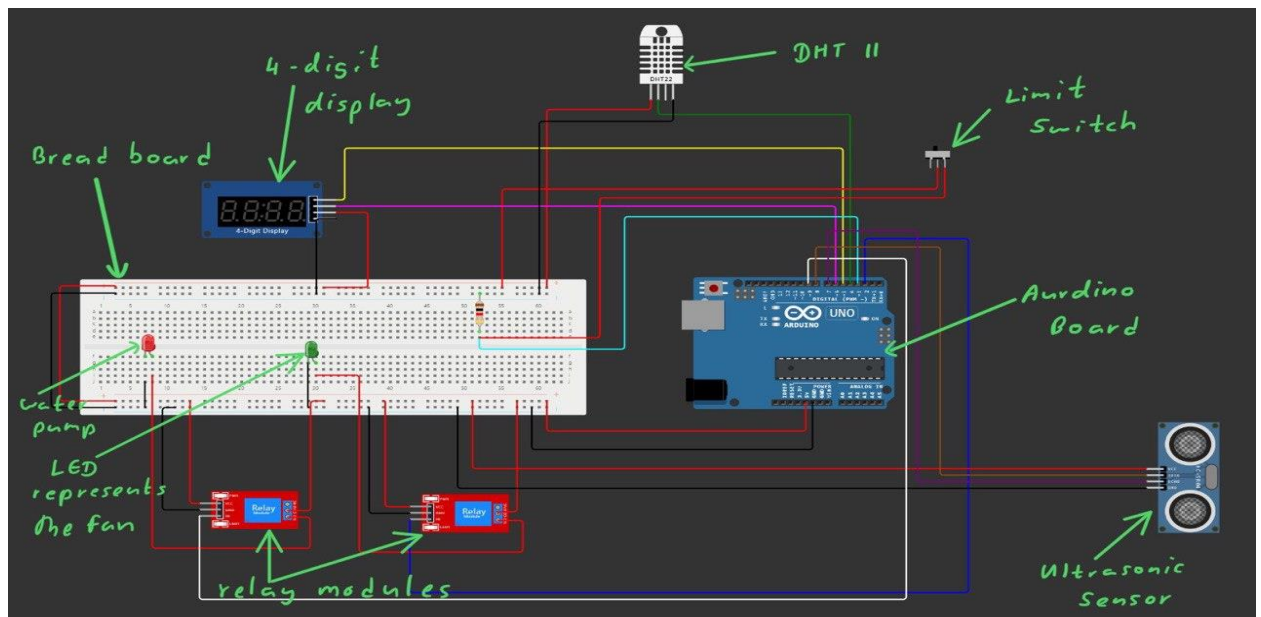
- Sheet metal
- Cotton wigs
- Rubber hose
- Binding wires

5.2 Procedure

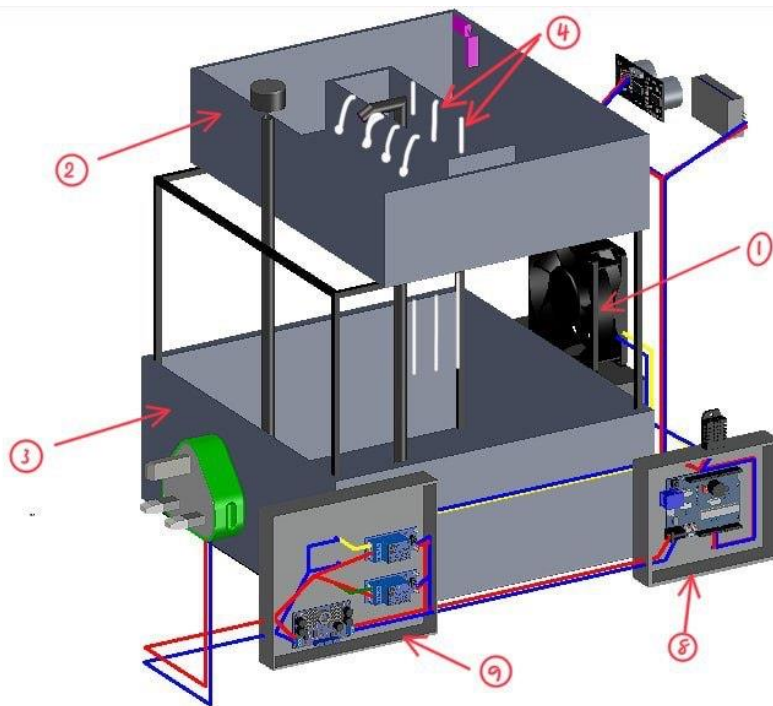
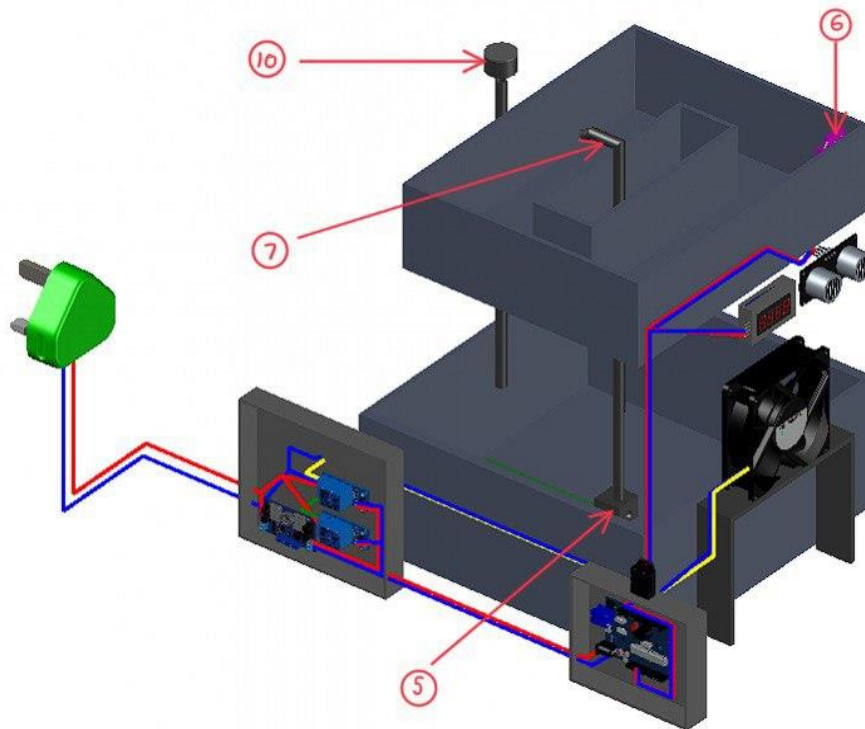
1. At first, we attached all the sheet metal panels with spot welding that was used as the frame of the system.
2. After that hardboard covering were cut into suitable sizes and make the outer box of the system
3. Then cotton wig plate was added to the top of the system.
4. Fan was glued to the body.
5. The water pump and the limit switch were set in necessary places with the help of a glue gun and nuts.
6. A sponge glued under the limit switch.
7. The Arduino uno board and the breadboard were attached to the box and then attached other components, humidity sensor, digital display, ultrasonic sensor, and relay modules.
8. Finally, Air ventilation holes were made for a proper air flow and a net was attached to the back of the body



5.3 The process of making circuit



5.4. Components



1. Fan
2. Water container 1
3. Water container 2
4. Cotton wicks
5. Small water pump.
6. Flouter switch / Limit switch
7. Pipe
8. DC unit
9. AC unit
10. Water inlet

5.5. Price list

component	Price(Rs)
12V DC Fan	800
Plug top	350
Humidity sensor	320
TT wires and binding wires	130
On/off rocker switch	50
0.5 inch hose	225
Glue sticks	40
Cotton wicks	240
Water pump	1200
Limit switch	800
2 Relay modules	300
DHT 11	330
Arduino board	2100
Ultra sonic sensor	310
4 digit display	400

5.6.TOOLS

1. Glue gun
2. Scissors
3. Screwdriver
4. Drilling machine
5. Paper cutter
6. Electric Bouth
7. plier

5.7.TECHNIQUES

1. Spot welding

6.Conclusion

In conclusion, our team's project aims to develop an innovative cooling system that addresses the challenge of controlling unnecessary electricity usage and humidity while providing cost-effective and highly efficient cooling. We recognize the limitations of inexpensive fans and expensive air conditioning machines and seek to bridge the gap by combining affordability with exceptional performance. By utilizing a sensor to detect proximity and movement, we can ensure that the system only operates when needed, saving electricity. Additionally, we incorporate natural humidity control using water. Throughout the design process, we prioritize sustainability, cost-effectiveness, suitable materials, ease of design, overall benefits, and safety.

7.Reference

- 1.Sharma, C. (2021) *Working Principle of Table Fan*, StudiosGuy. Available at:<https://studiousguy.com/working-principle-table-fan/> (Accessed: 08 July 2023).
2. *How does an air conditioner (AC) work?* (2022) Science ABC. Available at: <https://www.scienceabc.com/innovation/air-conditioner-ac-work.html> (Accessed: 08 July 2023).
3. *What is evaporative cooling and how does it work?* (no date) Oxycom. Available at: <https://www.oxy-com.com/what-is-evaporative-cooling> (Accessed: 08 July 2023).
- 4.*How does evaporative cooling work?: Portacool* (2022) Portacool Evaporative Coolers. Available at: <https://portacool.com/how-does-evaporative-cooling-work/> (Accessed: 08 July 2023).
5. Heatley, K. (2021) *The physics of keeping things cool*, MIT Physics. Available at: <https://physics.mit.edu/news/the-physics-of-keeping-things-cool/#:~:text=Putting%20something%20on%20ice%20to,and%20the%20latter%20to%20melt.> (Accessed: 08 July 2023).
6. Flylow (2021) *DT6-1A-01 I2C Arduino UNO code (seat occupancy detection)*, Arduino Forum. Available at: <https://forum.arduino.cc/t/dt6-1a-01-i2c-arduino-uno-code-seat-occupancy-detection/856945> (Accessed: 08 July 2023).
7. *Air conditioner efficiency* (no date) *Air Conditioner Efficiency | EGEE 102: Energy Conservation and Environmental Protection*. Available at: <https://www.e-education.psu.edu/egge102/node/2106> (Accessed: 08 July 2023).
- 8.*Efficient Air Conditioning Systems* (no date) *Efficient air conditioning systems | Climate Technology Centre & Network | Tue, 04/04/2017*. Available at: <https://www.ctc-n.org/technologies/efficient-air-conditioning-systems> (Accessed: 08 July 2023).
9. *Arduino - temperature sensor* (no date) *Online Courses and eBooks Library*. Available at: https://www.tutorialspoint.com/arduino/arduino_temperature_sensor.htm (Accessed: 08 July 2023).
10. (No date) *Basic concepts1*. Available at: <https://www.hec.usace.army.mil/confluence/hmsdocs/hmstrm/evaporation-and-transpiration/basic-concepts1#:~:text=Thermodynamics%20of%20Evaporation,to%20affect%20the%20state%20transition.> (Accessed: 08 July 2023).
11. *Best air conditioners price in Sri Lanka*: (no date) *Best Air Conditioners Price in Sri Lanka | - BuyAbans.com*. Available at: <https://buyabans.com/air-conditioners> (Accessed: 08 July 2023).

12. *Fan price in Sri Lanka - buy stand fan online* (no date) *Online Shopping Sri Lanka: Clothes, Electronics & Phones*. Available at: <https://www.daraz.lk/fans/> (Accessed: 08 July 2023).
13. *Best fans price in Sri Lanka*: (no date) *Best Fans Price in Sri Lanka* |- *BuyAbans.com*. Available at: <https://buyabans.com/home-appliances/fans> (Accessed: 08 July 2023).

8.APPENDICES

8.1.Appendix I: Alternative Conceptual Designs

DESIGN OF AN AUTOMATED COOLING FAN WHICH WE CAN USE
AS A SOLUTION TO THE HIGH COST FOR COOLING DEVICES



DESIGNED BY,
NAVINNA R. R
210410U
GROUP 04

1.Summary of the Problem

The problem with using a cooling device, such as an air conditioner, in a home is that it might result in a significant increase in electricity costs. This is since cooling systems need a lot of electrical power to operate, especially if they are used frequently or for extended periods of time. Limiting the use of cooling systems, purchasing energy efficient cooling units, enhancing insulation to keep heat from entering the house, or raising the thermostat are all ways to lower the cost of power.

2.Introduction the Concept

These fans are equipped with a thermostat that senses the temperature in the room and adjusts the fan's speed accordingly. When the temperature in the room reaches a desirable level, the fan turns off automatically to conserve energy and prevent the room from getting too cold. Similarly, when the temperature in the room increases beyond the desired level, the fan turns on automatically to circulate the air and cool down the room. This automatic on/off feature not only provides convenience but also helps to reduce energy consumption and save money on electricity bills.

3. Working Techniques

The fan that turns off automatically after the room cools down and turns on again when the room warms up is usually equipped with a thermostat. The thermostat is a device that can sense the temperature in the room and send signals to the fan to adjust its speed accordingly. When the temperature in the room reaches a comfortable level, the thermostat signals the fan to turn off automatically, conserving energy and preventing the room from getting too cold. On the other hand, when the temperature in the room increases beyond the desired level, the thermostat signals the fan to turn on automatically, circulating the air and cooling down the room. This automatic on/off feature not only provides convenience but also helps to reduce energy consumption and save money on electricity bills. Some fans may also come with additional features such as timers or remote controls, allowing users to further customize their cooling experience.

4. Making the Conceptual Design

4.1 Making the cooling part of the design.

- Cut 2 empty water bottles in half 1 in (2.5 cm) from the bottom.
- Drill a series of holes 0.25 inches apart above the cuts.
- Use string or wire to attach the bottles upside-down to the backside of the fan.
- Switch on your fan and place ice cubes inside the bottles.
- Add melted ice into a bowl.



Front view

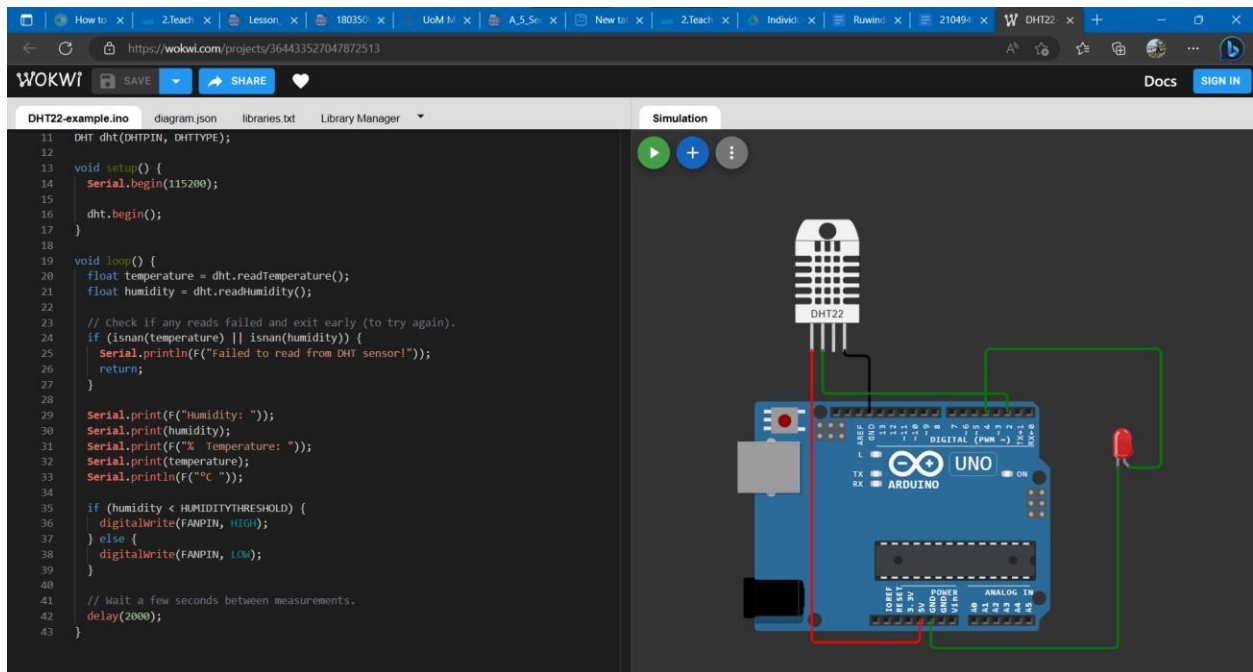
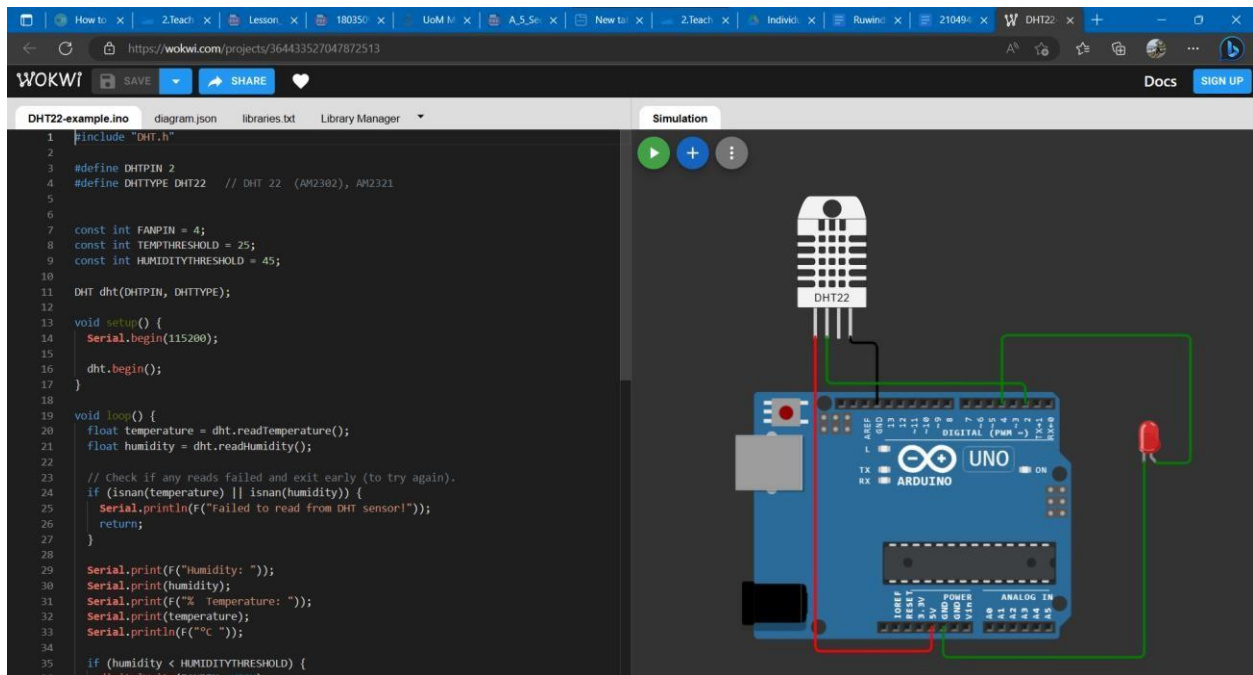


Side view



Back view

4.2 Making the sensor part of the design.



This is the Arduino program where the fan turns off automatically after the room cools down to comfortable level, and the fan turns on automatically when the room warms up again. People are usually comfortable at 45-percent relative humidity.

Here the LED represent the fan. It uses a DHT22 temperature and humidity sensor to read the current temperature and humidity, and then turns on a fan connected to digital pin 4 if the humidity level falls below a predefined threshold value of 45%.

Here's a breakdown of the code:

The program starts by including the DHT library and defining some constants. The DHT library provides functions for reading temperature and humidity from the DHT22 sensor, which is connected to pin 2 on the Arduino board. The program also defines a constant for the fan pin (4), and threshold values for humidity (45%) and temperature.

```
1  #include "DHT.h"
2
3  #define DHTPIN 2
4  #define DHTTYPE DHT22  // DHT 22 (AM2302), AM2321
5
6
7  const int FANPIN = 4;
8  const int TEMPTHRESHOLD = 25;
9  const int HUMIDITYTHRESHOLD = 45;
10
11  DHT dht(DHTPIN, DHTTYPE);
```

In the `setup()` function, the program starts serial communication at a baud rate of 115200 and initializes the DHT22 sensor.

```
12
13  void setup() {
14      Serial.begin(115200);
15
16      dht.begin();
17  }
```

In the `loop()` function, the program reads the temperature and humidity values from the DHT22 sensor using the “readTemperature()” and “readHumidity()” functions provided by the DHT library.

```

19 void loop() {
20     float temperature = dht.readTemperature();
21     float humidity = dht.readHumidity();
22

```

The program then checks if the temperature and humidity values are valid. If either of the values is NaN (not a number), it prints an error message to the serial monitor and exits the loop.

```

23     // Check if any reads failed and exit early (to try again).
24     if (isnan(temperature) || isnan(humidity)) {
25         Serial.println(F("Failed to read from DHT sensor!"));
26         return;
27     }
28

```

If the temperature and humidity values are valid, the program prints the values to the serial monitor.

```

29     Serial.print(F("Humidity: "));
30     Serial.print(humidity);
31     Serial.print(F("% Temperature: "));
32     Serial.print(temperature);
33     Serial.println(F("°C "));
34

```

The program then checks if the humidity value is below the predefined threshold value of 45%. If the humidity is below the threshold, it turns on the fan by setting the digital pin connected to the fan to HIGH. If the humidity is above the threshold, it turns off the fan by setting the digital pin to LOW.

```

35     if (humidity < HUMIDITYTHRESHOLD) {
36         digitalWrite(FANPIN, HIGH);
37     } else {
38         digitalWrite(FANPIN, LOW);
39     }
40

```

Finally, the program waits for 2 seconds before starting the loop again to allow time for the temperature and humidity to stabilize.

```

40
41 // Wait a few seconds between measurements.
42 delay(2000);
43 }

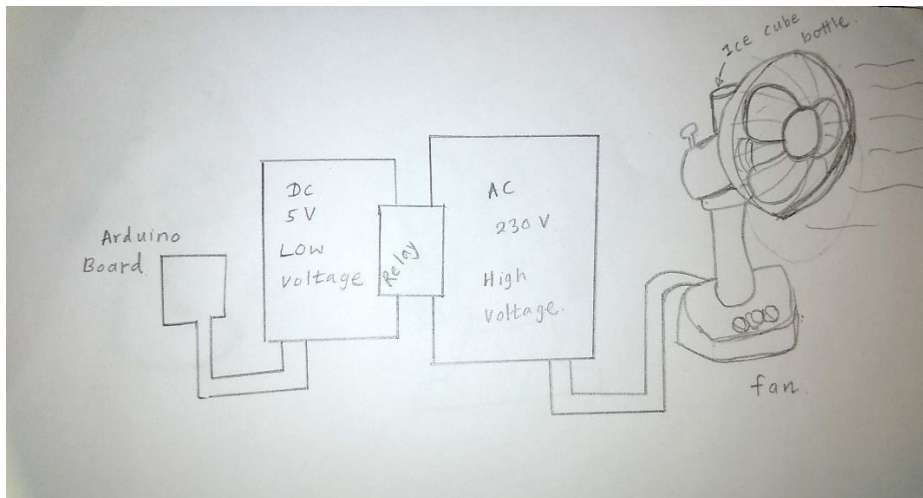
```

The process can be seen more clearly through this link.

https://drive.google.com/file/d/18_eD6Kv_AmKEWAaPyRz6twGR4YXBPnIz/view?usp=sharing

4.3 Attach sensor kit and setup the power supply

Since the fan is 230V, the fan is set to the Arduino board through a relay module.



5. Efficiency

The efficiency of the above program will depend on several factors such as the accuracy and precision of the humidity sensor, the performance of the fan, and the speed and reliability of the Arduino board.

The application should be effective in controlling the fan based on the surrounding temperature if the temperature sensor and fan are of high quality and the program is correctly developed. The program is made to turn on the fan when the temperature rises above a specific threshold and turn it off when the temperature drops below that threshold, which can assist in managing the environment's temperature and preventing overheating.

5.1 Energy

The power requirements of the Arduino board, the fan, and the temperature sensor, as well as the frequency of operation and the duration of time the fan is on, will all have an impact on how much energy is used by the mentioned software.

The software should consume relatively little energy if the Arduino board, temperature sensor, and fan are all low-power gadgets. The temperature sensor and fan should both demand little power, and the Arduino board may often be powered by a USB connection or a battery pack.

However, when the fan is running and using energy to cool the room, the energy usage would rise. The fan's power rating, the speed at which it is running, and how long it has been on will all affect how much energy it uses.

The program's overall energy usage will depend on a number of variables, so it's critical to optimize the system for energy efficiency by choosing low-power components, reducing frequency of operation and duration, and implementing power-saving features like sleep modes for the Arduino board and other components.

5.2 Cost

The equipment required for the project is,

- Arduino board
- Humidity sensor
- Fan
- Wires
- Relay module
- Discarded water bottles.

For this you can take a fan that currently used at home. Therefore, no money is spent on the fan. Also, no money is spent on disposable water bottles.

We have to spent money for Arduino board, humidity sensor, relay module and wires.

Arduino board – Rs 2475
Humidity sensor – Rs 800
Relay module - Rs 180
Wires – about Rs 50

According to the above data, only Rs 3500 will be spent for this project.



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5VDC 1 Way 1 Channel Relay Module (Transistor Version) (MD0486)



Name	5VDC 1 Way 1 Channel Relay Module (Transistor Version)
Code	MD0486
Price	Rs.180.00
In Stock	Yes
Package	MODULE

[MD0486 - 5VDC 1 Way 1 Channel Relay Module \(Transistor Version\) \(tronic.lk\)](http://tronic.lk)

DHT22 / AM2302 Digital Temperature And Humidity Sensor Module (MD0229)

Products

Name	DHT22 / AM2302 Digital Temperature And Humidity Sensor Module
Code	MD0229
Price	Rs.800.00
In Stock	No
Package	MODULE

[MD0229 - DHT22 / AM2302 Digital Temperature And Humidity Sensor Module \(tronic.lk\)](#)

6. Maintenance and safety in design

When using any program involving electronic devices, like the one mentioned before to control a fan based on the surrounding temperature, maintenance and safety must be given careful consideration.

Maintenance:

- Check the connections between the parts on a regular basis to make sure they are secure and working properly.
- Regularly check the temperature sensor to make sure it is still accurate and giving correct data.
- Check the fan blades for any wear or damage and replace them as necessary.
- Maintain a clean, dust-free environment in the system to guard against component damage.

Safety:

- Confirm the safety and compliance with applicable electrical codes and standards of the power supply going to the Arduino board and other components.

- To avoid overloading, short circuits, and other electrical risks, use the proper wiring and circuit protection.
- When working with the parts, keep to basic electrical safety precautions, such as keeping your hands away from any exposed wires and turning off the power before making any adjustments or repairs.
- Make sure the system is correctly grounded to avoid problems like electrical shocks.
- When handling the temperature sensor, be careful because it could get hot.

7. Ease of Design

Level of programming, electronics, and Arduino board working experience will determine how simple it is to design the circuit described above. Therefore, designing the software should be quite simple.

There may be a learning process involved in understanding the concepts and abilities required to develop the program if you are new to programming and electronics. However, there are a lot of internet resources, including tutorials, manuals, and discussion boards, that may help you learn these skills and get through any challenges.

The ease of designing the program will depend on your level of experience and familiarity with the necessary skills and tools. So anyone can create and carry out the software mentioned earlier with the correct resources and a willingness to learn.

8. Sustainability

The environmental impact of the project's components, the program's energy efficiency, and the program's potential to minimize energy use and greenhouse gas emissions are just a few of the variables that affect how sustainable the above-mentioned project is.

It's important to take environmental impact factored in when choosing project components, taking into consideration factors like the materials used, the energy needed to make them, and the alternatives for disposal at the end of their useful lives. The program's sustainability can be improved by selecting components that are constructed of environmentally friendly materials, have a low effect, and can be recycled or disposed of properly.

We use discarded water bottles in this project. So, it is somewhat eco-friendly.

The energy efficiency of the program is an essential part of its sustainability. The software can help to reduce energy consumption and related greenhouse gas emissions by being optimized to limit energy consumption, such as using low-power components, putting power-saving measures into place, and reducing the frequency and length of operation.

By lowering energy consumption of the fan can also benefit sustainability by lowering energy costs and the amount of grid electricity that is needed. This may aid in lowering the system's overall carbon impact.

**DESIGN OF A LOW-COST COOLING SYSTEM AS A
SOLUTION FOR THE HIGH COST OF COOLING DEVICES**

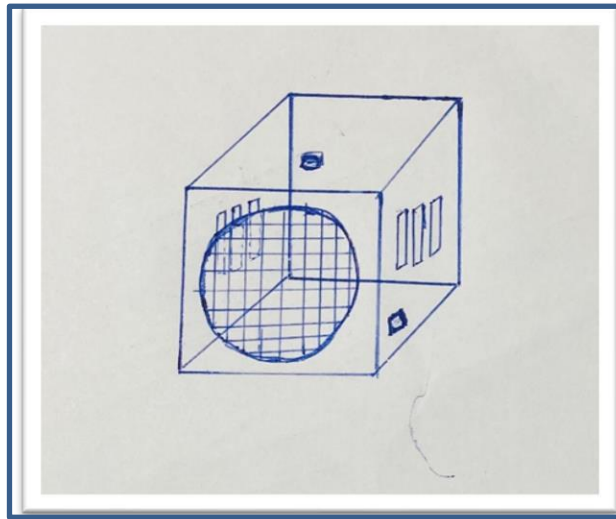
DESIGNED BY, PATHIRAGE SSK 210447M

1. Summary of the problem

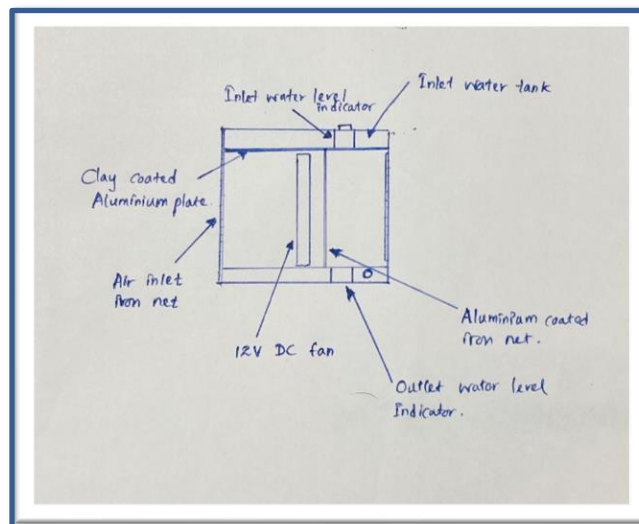
The high price of conventional air conditioning (AC) systems is a significant issue, restricting their use.

Installation, maintenance, and energy costs limit their accessibility. Designing an affordable alternative cooling system that offers equivalent performance and comfort while lowering costs and environmental impact is the task at hand.

2. Introduction to the concept



Outside View



Side View

3.Working Techniques

3.1 Clay Coated Aluminum Plate

The aluminum plate was chosen because it has good thermal conductivity, which means it can quickly absorb and dissipate heat. A layer of clay is applied to the surface of the aluminum plate. The clay serves as an insulator, which helps to slow down the rate at which the plate absorbs heat from the surrounding air. The plate is then cooled down by placing it in a refrigerator or freezer, or by running cold water over it. As the plate cools down, it absorbs heat from the surrounding air, causing the air to cool down as well. Once the plate has reached the desired temperature, it is mounted horizontally at the top of the fan housing, with the fan blades mounted directly underneath it. When the fan is turned on, it blows air over the cold plate, causing the air to be cooled down by the plate. As the air passes over the plate, the heat in the air is transferred to the plate, which absorbs it and dissipates it into the surrounding environment. The cooled air is then blown out of the fan housing and into the surrounding space, providing a refreshing breeze that helps to cool down the area.

3.2Aluminium coated iron net

An aluminum net was chosen because it has good thermal conductivity, which means it can quickly absorb and dissipate heat. The net should have a fine mesh design to ensure that it is able to trap small water droplets.

As the water flows through the net, it absorbs heat from the surrounding air, causing the air to cool down as well. The cooled air is then blown out of the fan housing and into the surrounding space, providing a refreshing breeze that helps to cool down the area. The water that has flowed through the net is collected at the bottom of the fan housing at the outlet water tank.

4.Making the conceptual design

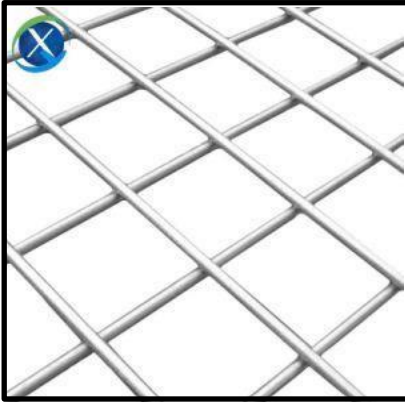
4.1. Making the outer shell

Here we can build the outer shell of the cooler using hard boards. In the front there will be an iron net that can be used to let air out according to preference.

At the top and the bottom there will be Aluminum tanks respectively for the inlet and outlet water that contains plastic caps to close and contains an indicator.

Moreover, there will be holes at the sides for better air ventilation and the power on/off button.

4.2. Setting up the equipment



Then inside the box we attach the 12V DC fan to the box.

Then to the back frame of the box we attach a iron net and then we apply some clay coat on the top of the inlet water tank. And then we attach the Aluminum coated iron net in front of the DC fan. And then air holes should be created in the side of the frame.

Finally, we connects the wires and the switches together as an appropriate manner.

5. Efficiency

5.1. Energy

The energy usage of the cooling fan setup will depend on the specific components used and the settings selected.

However, generally speaking, the use of a low-power fan and a chilled surface or water flow system can help to keep energy usage low.

Additionally, using an energy-efficient power supply can also help to minimize energy consumption. It is important to note that the energy usage may increase if a more powerful fan or a refrigeration unit is used to cool the surface or water. Overall, designing the system with energy efficiency in mind and selecting components with low power requirements can help to keep energy usage to a minimum.

5.2. Cost

Hard Board – Rs. 500

Aluminum Plates – Rs. 2500

Iron nets – Rs. 2000

12V cooling fan – Rs. 1000

Wires – Rs.200

Plastic Inlet caps and indicators – Rs. 1500

On/off switch – Rs. 400

Total cost of the product will vary from around 8500.

A design of a pipe line system to improve the air flow in a house with low power

DESIGNED BY,Perera G.D.D.C

NAme: Perera G.D.D.C

Index : 210458X

1.Summary of the Problem

The problem is that current cooling systems, such as air conditioners and refrigerators, have a significant impact on electricity bills. These systems require a large amount of energy to operate, especially during hot weather or when they are used frequently. This leads to increased electricity consumption and higher bills for consumers. The high energy usage of cooling systems also contributes to environmental issues, such as greenhouse gas emissions and climate change. Therefore, finding ways to improve the efficiency of cooling systems and reduce their impact on electricity bills is an important issue.

2.Introduction the Concept

A single pipeline system for in and out air can also be used to cool a house by providing a continuous flow of fresh air while removing warm, stale air.

3. Working Techniques

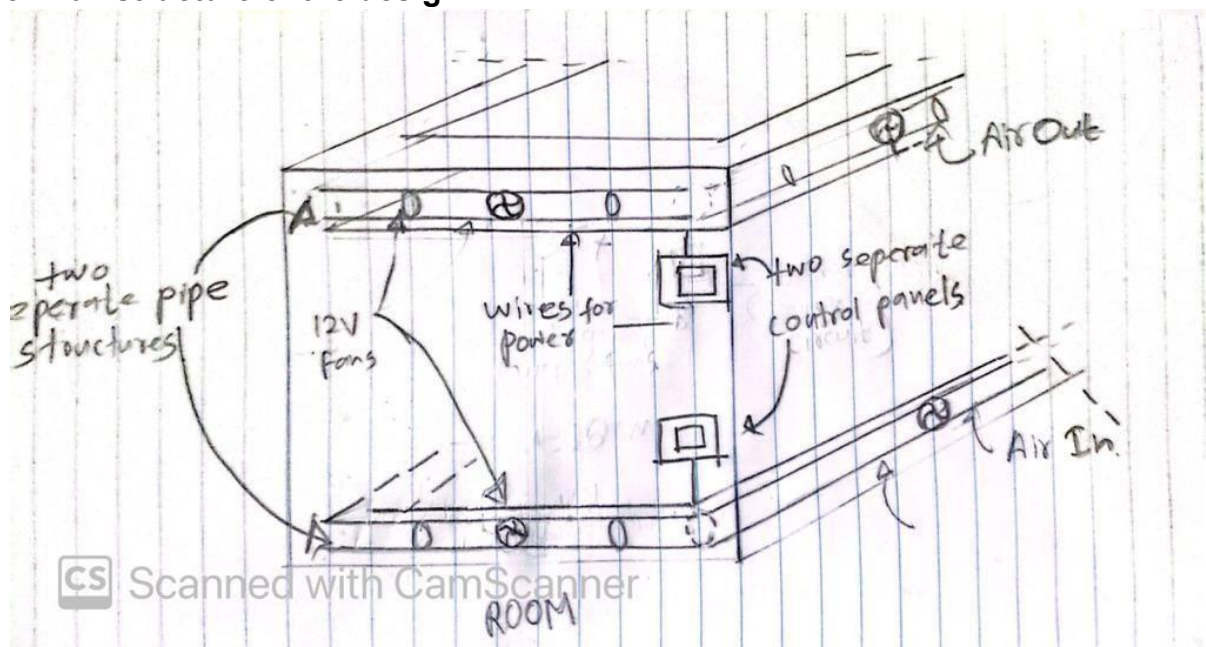
3.1 pipeline fan system

- The house is divided into two separate zones, with one zone located on the upper floor and the other on the lower floor.
- A set of pipelines is installed on the top wall of the house for the exhaust air, and another set of pipelines is installed on the bottom wall of the house for the intake air. The size and number of pipelines will depend on the size of the house and the amount of air required to cool it.
- Each pipeline is equipped with its own set of fans, with at least one fan located in each room or area of the zone. The fans are connected to a separate power supply and control module, which allows them to be turned on and off and to adjust their speed independently of the other fans.
- The exhaust pipelines are connected to a central air handling unit or a heat recovery ventilation system, which is responsible for removing warm, stale air from the house. The intake pipelines are connected to the same air handling unit, which is responsible for bringing in fresh air from outside the house.
- A separate control mechanism is used to adjust the speed and on/off status of each fan in each zone. This could be a manual control switch or a

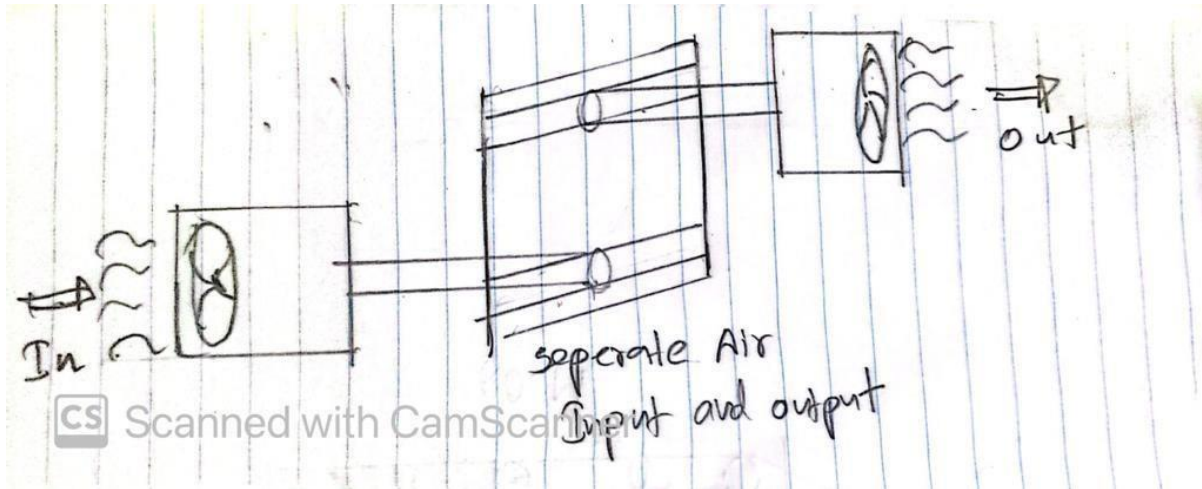
programmable controller that is connected to a computer or other control system.

- When the control mechanism sends a signal to a specific fan in a specific zone, that fan is turned on or off or its speed is adjusted based on the signal received. The other fans in the same zone are not affected by the signal.
- The system can be used to provide a continuous flow of fresh air to cool the house, with warm, stale air being removed from the house through the exhaust pipelines and cool, fresh air being brought in through the intake pipelines.
- Regular maintenance and inspection of the system is required to ensure that all fans and pipelines are functioning correctly and to identify any issues that may arise. This could include cleaning the fans and pipelines, checking for leaks or obstructions, and testing the control mechanism to ensure it is working correctly.
- Overall, this single pipeline system can be an effective and efficient way to cool a house by providing a constant supply of fresh air while removing warm, stale air.

3.2 full structure of the design



3.3 air in and out



3.4 Power supply to the device

When powering a multiple fan system, it's important to ensure that the power supply is adequate to meet the power requirements of all the fans in the system.

Each fan will have a rated voltage and current draw, which can be found in the fan's datasheet. To determine the total power requirements of the system, you can add up the power requirements of each fan.

The power requirements of the fans can be calculated using the following formula:

$$\text{Power} = \text{Voltage} \times \text{Current}$$

For example, if you have three 12V fans with a current draw of 0.2A each, the total power requirement of the system would be:

$$\text{Power} = 12\text{V} \times 0.2\text{A} \times 3 = 7.2\text{W}$$

Once you know the total power requirement of the system, you can choose a power supply that can provide at least that amount of power. It's generally a good idea to choose a power supply with a little bit of extra headroom to ensure that the fans have enough power to start up and run smoothly.

When selecting a power supply, make sure that the voltage rating of the power supply matches the rated voltage of the fans. You should also make sure that the power supply can provide enough current to meet the total current draw of the fans.

It's important to note that some fans may have different power requirements, depending on their speed and performance settings. In this case, it's a good idea to refer to the fan's datasheet to determine the power requirements for each specific setting, and choose a power supply that can handle the maximum power requirements of the system.

4. Making the pipe structure

To make a pipe structure for a multiple fan system, you will need to follow these steps:

- Determine the size and shape of the pipe structure needed to accommodate the number and size of the fans in your system. The pipe structure should be large enough to provide sufficient airflow and minimize turbulence, but not so large that it restricts the flow of air.
- Choose a suitable material for the pipe structure. Common materials include PVC, metal, and flexible ducting. Make sure that the material is durable, heat-resistant, and appropriate for your application.
- Cut the pipes to the required length using a saw or pipe cutter. If necessary, you can use connectors to join multiple sections of pipe together.
- Drill or cut holes in the pipe for the fans to be mounted. Make sure that the holes are the appropriate size for the fans and are evenly spaced.
- Install the fans into the holes in the pipe. Secure them in place using mounting screws or brackets.
- Connect the fans to the power supply and control module using wires. Make sure that the wires are securely connected and that the polarity is correct.
- Mount the pipe structure in the desired location using brackets or straps. Make sure that it is securely mounted and that there is sufficient clearance around the structure for proper airflow.
- Test the system to ensure that the fans are working correctly and providing the desired airflow. Adjust the fan speed and control settings as needed.

4.1 code to control fans

```
int fanPin1 = 9; // Connect fan 1 to pin 9 int fanPin2 = 10; //  
Connect fan 2 to pin 10 int speedPin = 11; // Connect speed  
control pin to pin 11 int speedValue = 0; // Set initial speed value  
to 0
```

```
void setup() { pinMode(fanPin1, OUTPUT);
```

```
    pinMode(fanPin2, OUTPUT); pinMode(speedPin,  
    OUTPUT);  
}
```

```

void loop() { // Read input from a sensor or user input to change the speed value
speedValue = analogRead(A0) / 4; // Map analog input from 0-1023 to 0-255

// Update the speed of the fans based on the speed value analogWrite(speedPin,
speedValue);

// Turn on/off the fans based on a boolean variable boolean fansOn = true;
// Change this value to turn on/off the fans if (fansOn) {
digitalWrite(fanPin1, HIGH); digitalWrite(fanPin2, HIGH); } else {
digitalWrite(fanPin1, LOW); digitalWrite(fanPin2, LOW);
}

delay(100); // Wait for a short period of time before updating again
}

```

4.2 setup the power supply

To set up the power supply for a multiple fan system, follow these steps:

- Determine the total power requirements of the system by adding up the power requirements of each fan. This can be calculated using the formula: $\text{Power} = \text{Voltage} \times \text{Current}$. Make sure to account for any variations in power requirements for different speed settings of the fans.
- Choose a power supply that can provide at least the total power requirement of the system. It's generally a good idea to choose a power supply with a little bit of extra headroom to ensure that the fans have enough power to start up and run smoothly. Make sure that the voltage rating of the power supply matches the rated voltage of the fans.
- Connect the power supply to a power input port on a relay or transistor module, which will be used to control the fans. The power input port will typically be labeled with a positive (+) and negative (-) symbol.
- Connect the power output port of the relay or transistor module to the positive (+) input port of each fan. The negative (-) input port of each fan should be connected directly to the negative (-) output of the power supply.
- Connect the control pins of the relay or transistor module to the microcontroller or single-board computer that will be used to control the fans. The control pins will typically be labeled with a signal (S) symbol and a positive (+) and negative (-) symbol.

- Write code to control the fans using the microcontroller or single-board computer. This code should include commands to turn the fans on and off and adjust their speed using PWM (Pulse Width Modulation) signals.
- Test the system to ensure that the fans are receiving power and responding to the control signals correctly. If the fans are not responding correctly, check your wiring and code for any errors.

5. Efficiency

5.1 Energy

The energy efficiency of a connected pipeline fan system in a house depends on several factors, such as the size and layout of the house, the number of rooms and their ventilation needs, the type and quality of the fan system, and the insulation and airtightness of the house.

Here are some tips to improve the energy efficiency of a connected pipeline fan system in a house:

Choose the right fan system: Select a fan system that is appropriate for the size and ventilation needs of your house. Consider installing an energy-efficient model that has a high efficiency rating and uses less electricity.

Use variable speed controls: Install a variable speed control on the fan system, which allows you to adjust the speed of the fan based on the ventilation needs of each room. This can help save energy by reducing the amount of air that is circulated when it's not needed.

Ensure proper insulation and airtightness: Insulate your home and seal any gaps or leaks to prevent air from escaping or entering the house unnecessarily. This can help reduce the amount of energy needed to heat or cool your home, and reduce the workload on the fan system.

Use programmable thermostats: Set up a programmable thermostat to control the temperature of your home. This can help you optimize the operation of the fan system and save energy by reducing heating and cooling when you don't need it.

Perform regular maintenance: Keep your fan system clean and well-maintained to ensure that it operates efficiently. Check the air filters regularly and replace them as needed to maintain proper airflow and prevent the system from working harder than necessary.

Overall, taking these steps can help you improve the energy efficiency of your connected pipeline fan system and reduce your energy bills while keeping your home comfortable and healthy.

5.2 Cost

The cost efficiency of a connected pipeline fan system in a house depends on several factors, such as the cost of the fan system, the energy consumption of the system, and the savings achieved through improved energy efficiency.

Here are some tips to improve the cost efficiency of a connected pipeline fan system in a house:

Choose an energy-efficient fan system: Select a fan system that has a high energy efficiency rating and uses less electricity. While an energy-efficient model may cost more upfront, it can provide significant savings in the long run by reducing your energy bills.

Use variable speed controls: Install a variable speed control on the fan system, which allows you to adjust the speed of the fan based on the ventilation needs of each room. This can help save energy and reduce operating costs by reducing the amount of air that is circulated when it's not needed.

Optimize the ventilation system: Ensure that the ventilation system is designed and installed properly to optimize its performance. This includes sizing the system correctly and installing the ductwork properly to minimize resistance to airflow and reduce energy consumption.

Ensure proper insulation and airtightness: Insulate your home and seal any gaps or leaks to prevent air from escaping or entering the house unnecessarily. This can help reduce the amount of energy needed to heat or cool your home, and reduce the workload on the fan system.

Perform regular maintenance: Keep your fan system clean and well-maintained to ensure that it operates efficiently. This includes cleaning or replacing air filters regularly, checking ductwork for leaks, and ensuring that the system is balanced properly.

Overall, taking these steps can help you improve the cost efficiency of your connected pipeline fan system and reduce your energy bills while keeping your home comfortable and healthy. It's important to consider the upfront cost of the system and the potential savings over time when making a decision about which system to install.

6. Maintenance and safety in design

Maintenance and safety are important considerations when designing a connected pipeline fan system in a house. Proper design, installation, and maintenance of the system can help ensure that it operates efficiently and safely.

Here are some tips for maintenance and safety in the design of a connected pipeline fan system in a house:

Use high-quality materials: Choose high-quality materials for the ductwork, such as galvanized steel, aluminum, or fiberglass, to ensure that they are durable and safe. Avoid using materials that can deteriorate or corrode over time, such as plastic or cloth ductwork.

Install fire dampers and smoke detectors: Install fire dampers in the ductwork to prevent the spread of fire and smoke through the system. Also, install smoke detectors throughout the house to alert occupants in case of a fire.

Ensure proper ventilation: Proper ventilation is essential for the health and safety of occupants. Ensure that the system is designed and installed properly to provide adequate ventilation and airflow throughout the house.

Perform regular maintenance: Regular maintenance is critical to ensure that the system operates efficiently and safely. This includes cleaning or replacing air filters regularly, checking ductwork for leaks or damage, and ensuring that the system is balanced properly.

Provide access for maintenance: Ensure that the system is designed to provide access for maintenance and repair. This includes providing access panels or hatches for ductwork and ensuring that the fan unit is easily accessible for maintenance.

Overall, it's important to prioritize maintenance and safety in the design of a connected pipeline fan system in a house. Proper design, installation, and maintenance can help ensure that the system operates efficiently, reliably, and safely.

7. Ease of Design

The ease of designing a connected pipeline fan system in a house can depend on several factors, such as the size and layout of the house, the number of rooms and their ventilation needs, and the available space for installing ductwork and fans.

Here are some tips to make the design process easier for a connected pipeline fan system in a house:

Conduct a ventilation assessment: Before designing the system, conduct a ventilation assessment to determine the ventilation needs of each room. This can help you determine the appropriate size and type of fan system needed for each room.

Plan the layout: Plan the layout of the ductwork and fans to ensure that they are installed properly and efficiently. Consider factors such as the location of the fans, the size and type of ductwork needed, and the routing of the ductwork to ensure that airflow is optimized and resistance is minimized.

Use software tools: Use software tools to help you design and model the system. There are many software tools available that can help you model airflow and optimize the design of the system, making it easier to identify potential problems and optimize the design for efficiency and performance.

Consider ease of maintenance: Consider ease of maintenance when designing the system. Ensure that the system is designed to provide easy access for maintenance and repair, such as by providing access panels or hatches for ductwork and ensuring that the fan unit is easily accessible.

Consult with experts: Consult with HVAC experts to help you design the system. They can provide valuable advice and guidance on designing an efficient and effective connected pipeline fan system that meets your specific needs and requirements.

Overall, designing a connected pipeline fan system in a house can be made easier by conducting a ventilation assessment, planning the layout, using software tools, considering ease of maintenance, and consulting with experts. By following these tips, you can ensure that your system is designed properly, operates efficiently, and provides adequate ventilation and comfort for your home.

8. Sustainability

A connected pipeline fan system in a house can be a sustainable option for maintaining good indoor air quality and reducing energy consumption. Such a system typically consists of a network of ducts that connect different rooms in the house to a central fan that circulates air throughout the building.

The sustainability of this system depends on several factors, including the materials used in the construction of the ducts and fan, the energy efficiency of the fan, and the maintenance and cleaning of the system.

To ensure the sustainability of a connected pipeline fan system in a house, it is important to use high-quality, durable materials such as stainless steel or aluminum for the ducts, which will not degrade over time and will resist corrosion and damage from moisture and pests.

In addition, the fan should be designed to be energy-efficient, using a low-power motor and energy-saving controls to minimize electricity consumption. Regular maintenance and cleaning of the system are also critical to ensure optimal performance and prevent the buildup of dust, mold, and other contaminants that can affect indoor air quality and increase energy consumption.

Finally, it is important to consider the overall sustainability of the system, including its impact on the environment and the health and well-being of occupants. This may include using eco-friendly materials, selecting a fan with low noise levels, and ensuring that the system is properly sized and installed to provide adequate ventilation without wasting energy.

DESIGN OF AN TABLE COOLER WHICH WE CAN USE AS A SOLUTION TO THE HIGH COST FOR COOLING DEVICES

DESIGNED BY, PREMAKUMARA H.P.S.
210494D.

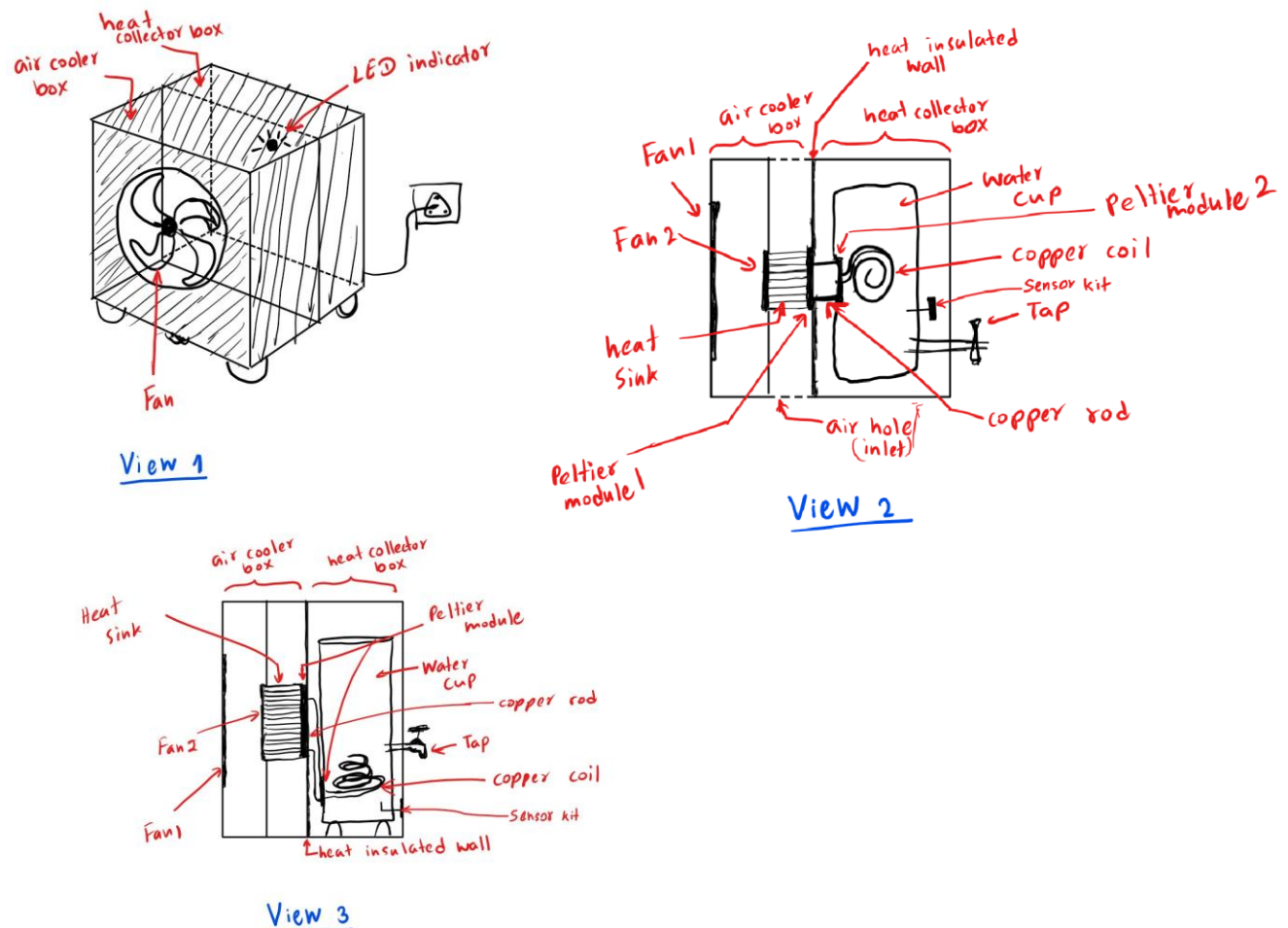
1.Summary of the Problem

The problem at hand is the need for a cooling device that is affordable, energy-efficient, and effective at cooling indoor spaces. Many existing cooling systems on the market can be expensive to purchase and operate, and may not be accessible to all customers.

In addition, there is a growing need for more sustainable cooling solutions that minimize energy consumption and environmental impact. Therefore, there is a demand for a cooling device that can provide a balance of affordability, efficiency, and effectiveness, while also being environmentally friendly.

2.Introduction the Concept

The rising demand for cooling devices has led to a significant increase in their cost, making them inaccessible to many people. To address this issue, I propose a new cooling device called the "**Heat Collector**" which is designed to provide affordable cooling and hot water supply.



The Heat Collector comprises two main features:

1. Cooling the air in front of the device

2. Generating hot water for drinking or other purposes The device consists of

two parts:

1. The air cooler box
2. The heat collector box.

The heat collector box absorbs some heat from the input room air, and then outputs cool air back into the room. At the same time, it transfers the absorbed heat energy to a water cup, which can be used to obtain hot water. The outside of the heat collector box features a red light that indicates the temperature in the water cup. When the light flashes, it is time to collect the hot water from the box. Afterward, one can add water at room temperature to the cup, and the cycle repeats as the device is used.

The Heat Collector provides a cost-effective and sustainable alternative to traditional cooling devices that can be inaccessible to many. By utilizing waste heat energy to generate hot water, the device offers an additional benefit, making it a versatile and useful appliance for customers.

3. Working Techniques

This device has two main parts, as I mentioned. Let's see how they work.

3.1 The Peltier module

The main component of this unit is the Peltier module.

A Peltier module is a device that uses the Peltier effect to create a heat flux at the junction of two different types of materials. The Peltier effect is one of the three phenomena that make up the thermoelectric effect, which is the conversion of temperature differences into electric voltage and vice versa.

A Peltier module consists of two copper substrates and many thermoelectric elements arranged in tandem and sandwiched between the substrates. The thermoelectric elements are made of two types of semiconductors: p-type and n-type. When a DC electric current flows through the module, it brings heat from one side to the other, so that one side gets cooler while the other gets hotter. This allows the module to have both "warming" and "cooling" effects depending on the direction of the current.

The main advantages of a Peltier module are its lack of moving parts or circulating liquid, long life, invulnerability to leaks, small size, and flexible shape. The main disadvantages are its high cost for a given cooling capacity and poor power efficiency (a low coefficient of performance or COP).

3.2 The Air-Cooling box

This is the main component of our device. It cools the room air and below is a top view of the air-cooling box.

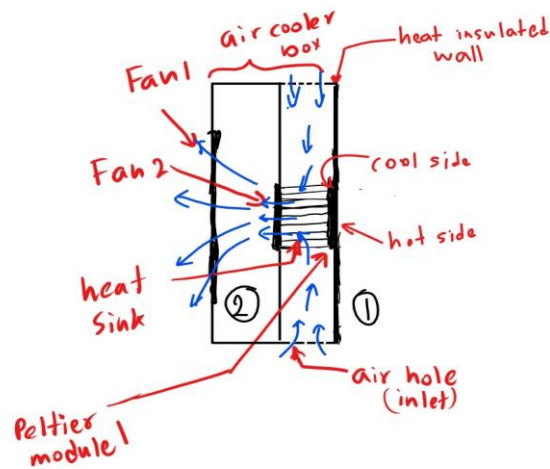


Figure 1

This part consists of two sections. The Peltier module is positioned in the middle of the heat-insulated wall, as shown in Figure 1. The heat sink is placed on the cool plate of the Peltier module, and a small cooling fan (fan 1) is attached to the top of the heat sink. When the fan is turned on, hot air from the room is drawn into section 1 of the cooling box through the air holes. The fan causes the drawn air to flow through the heat sink and into section 2, as indicated by the blue arrows in Figure 1. Fan 2 then pushes the cooled air out of the unit and into the room.

3.3 The heat collector box

This is the other component of my device. It is designed to collect wasted energy and convert it for use in something useful. Below is a top view of the heat collector box.

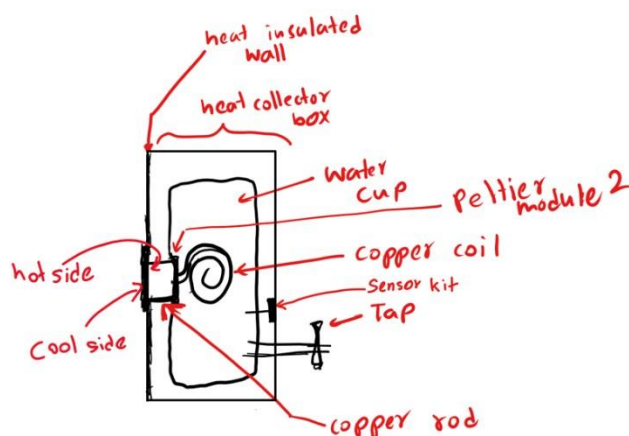


Figure 2

In this unit, we maintain waste energy. The Peltier module transfers the absorbed heat energy from the room air to this unit using electrical energy. However, this process also generates some heat energy, which accumulates near the hot plate of the Peltier module. We need to remove this heat

energy, but if we release it to the environment, we will lose a significant amount of energy. To solve this problem, I designed a water cup in this unit to release that heat energy to the water. We can then use this heat energy to boil water.

The start cross-section of a copper rod connects to the hot plate of the Peltier module which attached on the insulated wall and at the end of the copper rod, I attached another Peltier module, which operates at low voltage to facilitate heat transfer from the hot plate of the first Peltier module to the water cup. The hot plate of the second Peltier module is connected to the water cup and releases the heat energy to the water. To simplify this process, I attached a copper coil to this hot plate (see figure 2).

There is also a sensor unit in this section that informs the user when it's time to remove the water from the cup (that indicates water is hotter). If the user doesn't remove the water, the cooling device will automatically turn off. There is a tap for getting hot water from the device.

3.4 Both units work together

Both the air-cooling box and the heat collector box work together to create an efficient and sustainable cooling system. As hot air from the room is drawn into the air-cooling box through the air holes, the Peltier module which placed on the center of the heat insulated wall, cools the air by transferring the absorbed heat energy to the heat collector box. The heat collector box then converts this wasted energy into something useful by heating water in the water cup, which can then be used for various purposes such as making tea or coffee.

The two boxes are connected through the copper rod and second Peltier module in the heat collector box, which facilitate heat transfer from the hot plate of the first Peltier module to the water cup. As a result, the cooling device not only cools the room air but also provides a source of hot water.

The device is designed to operate seamlessly, with a sensor unit in the heat collector box informing the user when the water is ready to be used. If the water is not removed in time, the device will automatically turn off to prevent overheating.

Overall, the combination of the air-cooling box and heat collector box allows for a sustainable and energy-efficient cooling system that not only cools the room air but also provides a useful source of hot water.

3.5 Power supply to the device

The cooling device requires several components to operate, including two fans, two Peltier modules, and a sensor kit. Each of these components requires a separate power supply to function correctly. The fans are both maintained at 12V DC, with one fan placed on top of the heat sink and the other on the cooling box to flow out the cooled air.

The Peltier modules also require 12V DC to function properly, as this voltage is required to create the heat flux at the junction of the semiconductors. Additionally, the sensor kit requires power to operate, which is provided by an Arduino board and relay that switches the main power supply. The Arduino board and relay work at 5V DC.

To simplify the power supply to the device, only one power supply of 12V DC is needed. Therefore, the decision was made to use a 12V 5A power adapter to provide power to all the components. This ensures that all components are receiving the correct voltage and amperage needed to operate efficiently.

Overall, the power supply to the device is designed to be simple and efficient, with only one power adapter required to power all the necessary components. This allows for easier installation and maintenance of the device, while ensuring that all components are receiving the correct power supply to operate efficiently.

Making the Conceptual Design:

Before starting the actual construction process, it's crucial to create a conceptual design of the device. This design will help to identify the necessary materials, tools, and components required for building the device. The following steps will guide you through the process of creating a conceptual design.

Step 1: Identify the purpose of the device - The first step in creating a conceptual design is to identify the purpose of the device. For instance, if I am building a beverage cooler, I need to determine the type of beverages that the cooler will store.

Step 2: Determine the size of the device - After identifying the purpose of the device, the next step is to determine the size of the device. This will depend on the number of beverages I want to store and the available space.

Step 3: Draw a rough sketch of the device - Once I have identified the purpose and size of the device, draw a rough sketch of the device. This will help me to visualize the final product and make any necessary changes before starting the construction process.

Step 4: Identify the necessary components - Based on the rough sketch, identify the necessary components required for building the device. This will include the Peltier module, heat sink, cooling fans, copper rods, water cup, and sensor kit.

4.1 Making the main box of the device:

The first step in building the device is to make the main box that will house all the components. Follow these steps to make the main box.

Step 1: Measure and cut the wooden sheets - Measure and cut the wooden sheets according to the dimensions of the rough sketch.

Step 2: Join the wooden sheets - Join the wooden sheets using nails and screws to create the main box.

Step 3: Sand and paint the box - Sand the box to remove any rough edges and paint it with your preferred color.

4.2 Attach the Peltier module, heat sink, and cooling fans:

The next step is to attach the Peltier module, heat sink, and cooling fans to the main box. Follow these steps to attach the components.

Step 1: Apply thermal paste - Apply thermal paste on the cool side of the Peltier module.

Step 2: Attach the heat sink - Attach the heat sink on the cool side of the Peltier module.

Step 3: Attach the cooling fans - Attach the cooling fans on the heat sink to flow the room air through heat sink to absorb heat by the Peltier module.

4.3 Attach the copper rod to the hot side of the Peltier module:

The copper rod is used to transfer heat from the Peltier module to the water cup. Follow these steps to attach the copper rod.

Step 1: Cut the copper rod - Cut the copper rod to the required length.

Step 2: Bend the copper rod - Bend the copper rod at a 90-degree angle.

Step 3: Attach the copper rod - Attach the copper rod to the hot side of the Peltier module using thermal paste.

4.4 Water cup connect with copper rods:

The water cup will be attached to the copper rods. Follow these steps to connect the water cup with the copper rods.

Step 1: Drill holes in the water cup - Drill two holes in the water cup to insert the copper rods.

Step 2: Insert the copper rods - Insert the copper rods into the drilled holes.

Step 3: Secure the water cup - Secure the water cup using screws.

4.5 Attach sensor kit:

Next, I need to attach the temperature sensor kit to my device. This kit will allow you to monitor the temperature of the water in real-time. I can place the sensor in the water cup or attach it to the copper rod. Follow the instructions provided with the kit to ensure proper installation.

4.6 Setup the power supply:

Once I have assembled all the components, I need to connect the device to a power supply. I can use a 12-volt 5A power adapter to power the Peltier module and cooling fans. Make sure to connect the positive and negative terminals correctly. I can also add an on/off switch to control the power supply.

Once I have completed all the steps, my designed is cooler device is ready to use. Simply add water to the cup and turn on the device. The Peltier module will start cooling the water, and the fans will circulate the air to improve the cooling efficiency. The temperature sensor kit will allow you to monitor the temperature of the water and ensure that it is at the desired level.

5.Efficiency:

Efficiency is an important factor to consider when evaluating the performance of the device. The efficiency of the device can be measured in terms of its energy consumption, cost, and maintenance requirements.

5.1 Energy:

The energy efficiency of the device can be calculated by using the following equation:

$$\text{Efficiency} = \text{Output energy} / \text{Input energy}$$

Output energy is the amount of cooled air by the device, and Input energy is the amount of energy consumed by the device.

To calculate the output energy, you can use the following equation:

$$\text{Output energy} = \text{rate of Mass of inputted air} \times \text{Specific heat of air at constant pressure} \times \text{Temperature difference} \times \text{Time}$$

Where, Mass of ice produced is the amount of ice produced by the device, Specific heat of water is the amount of energy required to raise the temperature of water by 1 degree Celsius, and Temperature difference is the difference between the initial temperature of the water and the final temperature of the ice.

To calculate the input energy, we can use the following equation:

$$\text{Input energy} = \text{Power consumption} \times \text{Time}$$

Where, Power consumption is the amount of power consumed by the device, and Time is the duration for which the device is in operation.

Approximately input power is 60W.

Approximately output power is 9W.

Therefore efficiency is 15%.

5.2 Cost:

Equipment	Cost
1.peltier module x 2	Rs.1700x2
2.heat sink cooler	Rs.1000
3.cooling fan	Rs.1600

4.Material for box	Rs.2000
5.others	Rs.1000
Total	Rs.9000

5.3 Maintenance:

This is a simple device. Therefore maintenance of that very easy.

DESIGN OF A PORTABLE COOLING FAN THAT CAN
CONSUME LOW ENERGY

DESIGNED BY,
PREMARATHNE O. D
210496K

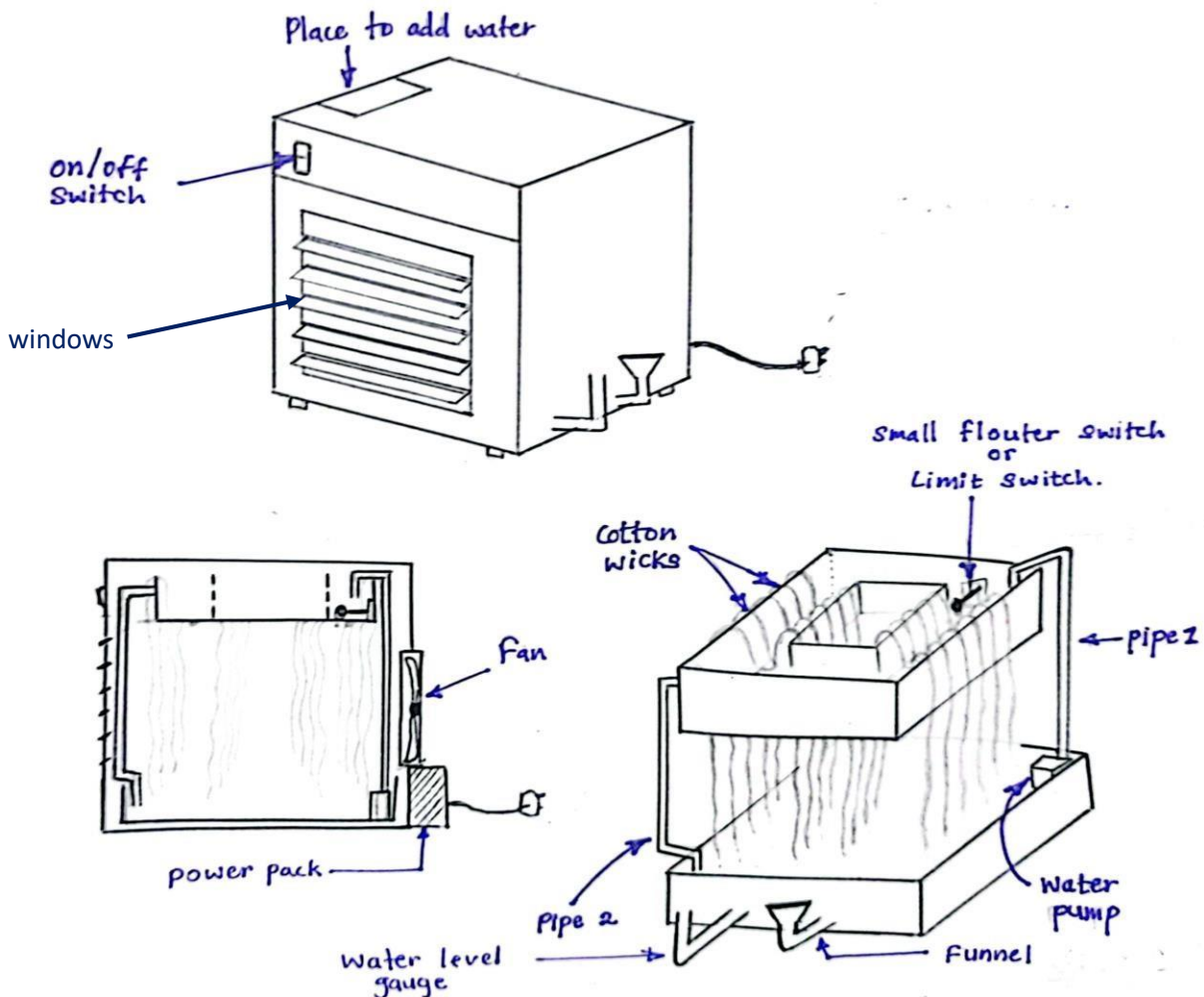
1.Summary of the Problem

There is a need of portable cooling device which is affordable and more energy efficient.

Fans and air conditioners are commonly used in indoor cooling. Air conditioners usually cool down the area efficiently but they are highly expensive and also highly affected to the electricity bill which consumes more current. Fans are also used for cooling but it can't cool down the area as air conditioners since they rotate the same air in the area. If there is a fan (ceiling fan) or a AC far from the working table, then the effectiveness of the cooling system is low.

Therefore, there is a requirement of a portable cooling system in affordable price and consumes low current to work as well as environmental friendly.

2.Introduction the Concept



3. Description about the portable cooling device

- **Fan** - It is used a 12V fan which can work from the battery current. This is used to flow the air through the wet wicks and reduce the temperature of the air in the area.
- **Water container 1** - It is made of metal sheets and shaped like a frame which is shown in the above figure. Its shape helps to hang the cotton wicks. Water we put in to the device at the start is stored in this container.
- **Water container 2** - This is also a metal container. This is used for store the water which flows through the cotton wicks.
- **Cotton wicks** - These are used to leak water slowly from the container 1. They are placed as shown in above figures.
- **Small water pump** - This is placed in the bottom container (container 2). It is used pump the water from the bottom container to the container 1.
- **Small flouter switch / Limit switch** - This is placed in the container 1. It is used to switch on the water pump when the water level goes down after certain level.
- **Pipe 1** - It is used to carry water from the container 2 to the container 1.
- **Pipe 2** - It is used to carry excess water in the container 1.
- **Power pack** - It is used to generate battery current to operate the fan, from the direct current.
- **Water level gauge** – To check the water level in container 2.
- **Funnel** – To enter water if the water level is very low in container 2.

3.1 Power supply to the device

The device can be connected to the domestic current via a wall outlet. Water pump works with the direct current(230V). Since the fan operates with battery current, it is used a power pack or a AC DC transformer to generate required battery current from the direct current.

4. Working technique

First we have to put water into the container 1. Then it starts to leak water from container 1 through the cotton wicks. When we turn on the switch, fan starts on and air flows through the wet cotton wicks. It cools the air and increases the humidity in the air as the air flows through the wet cotton wicks.

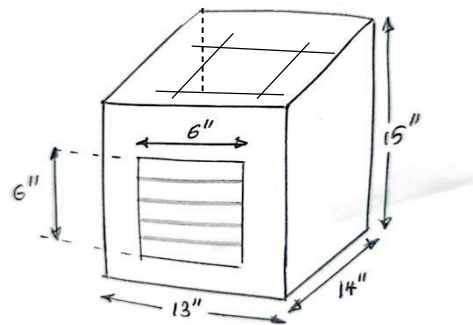
If the water level drops below a certain level after a few hours, the limit switch will automatically turn on. Then the water pump starts and brings water from container 2 through pipe 1 to container 1.

If there is a fault in the limit switch, then the water pump will work continuously and there will be an overflow in the container 1. To avoid that pipe 2 is used to carry excess water in the container 1.

4. Making the Conceptual Design

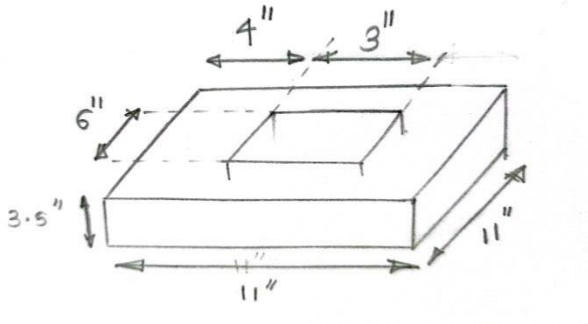
4.1 Making the main box of the device

Main box can be made of wood /metal or thick board with following dimensions. It is needed to leave space area in the front of the box to flow the cooled air. And also we have to make a frame inside the box to place container 1 properly.

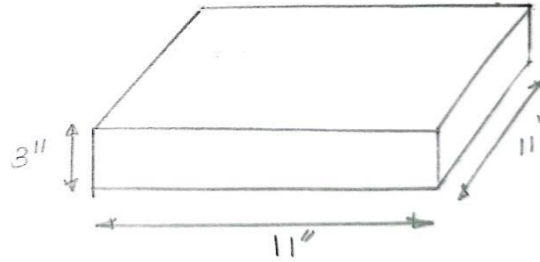


4.2 setup the water containers and cotton wicks

Water containers are made of metal sheets. The dimensions are shown in the below figures.

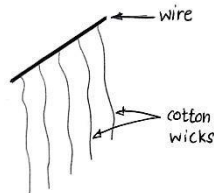


Container 1



Container 2

Cotton wicks are connected to a metal wire. Then it can be placed in the container 1 easily.



5. Efficiency

5.1 Energy

This cooling device consumes 15W – 20W power while normal exhaust fans and table fans consume 35W – 50 W. This cooling device can be work continuously 5 to 6 hours using nearly 1 liter of water.

5.2 Cost

	<u>Cost(Rs)</u>
1. Fan (12v)	1500
2. Cotton wicks	500
3. Water pump	1500

4. Limit switch	200
5. Power pack (AC/DC transformer)	2000
6. Water pipes	500

Approximately maximum price = Rs 8500 – Rs 9000

6. Maintenance and safety in design

- It is very easy to maintain this cooling device. To clean the cotton wicks and containers, container 1 and container 2 can be removed and fixed again easily.
- We can change the direction of the air flow by touching the windows in the cooling device.
- When we don't use the cooling device it is needed to close the funnel and the water level gauge to prevent mosquito breeding.

7.Sustainability

In this portable cooling device water is the only thing which is used to the cooling process. To decrease the temperature much more it can be added ice cubes into the container 1.

This portable device is an eco-friendly product as this device can be made of reusable materials as well as there is no use of chemicals to maintain the device and it doesn't release anything harmful to humans and the environment.

DESIGN OF A FAN THAT DOESN'T SPEND ENERGY FOR THE COOLING EFFECT

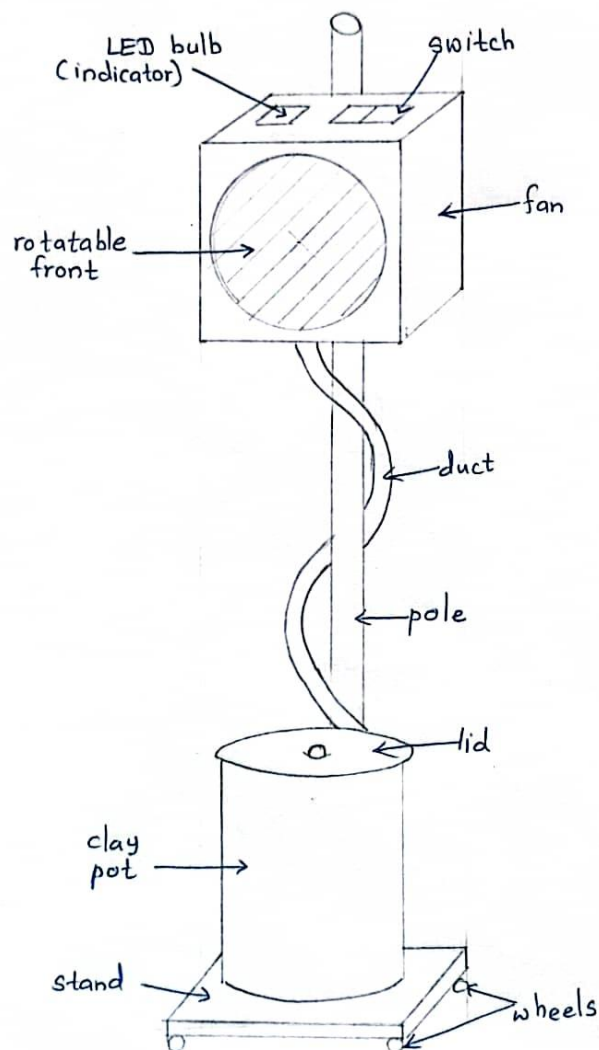
DESIGNED BY,
RAJAPAKSHA W.R.A.K.H.
210509G

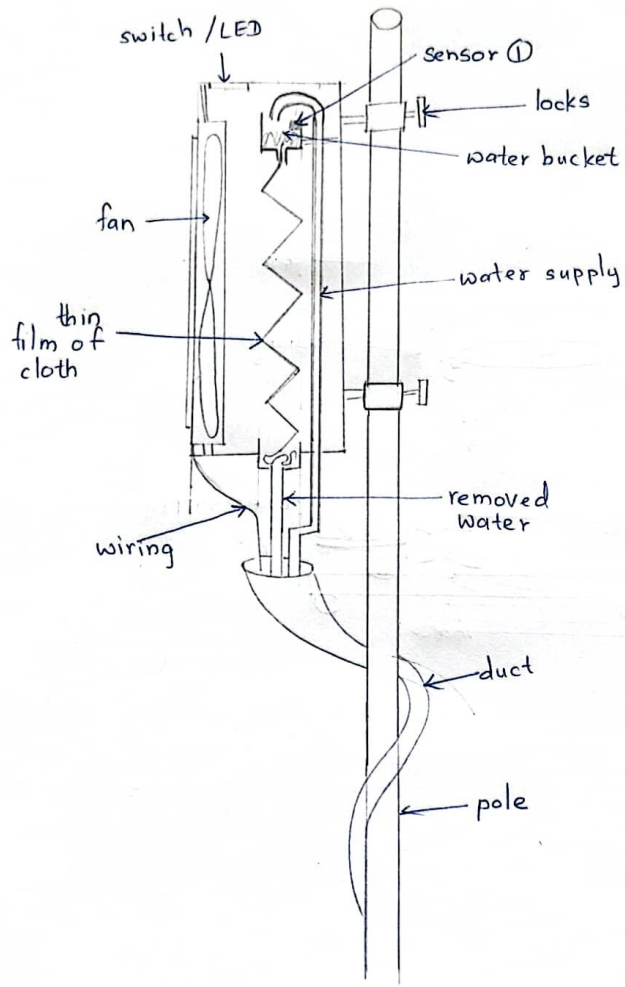
1.Summary of the Problem

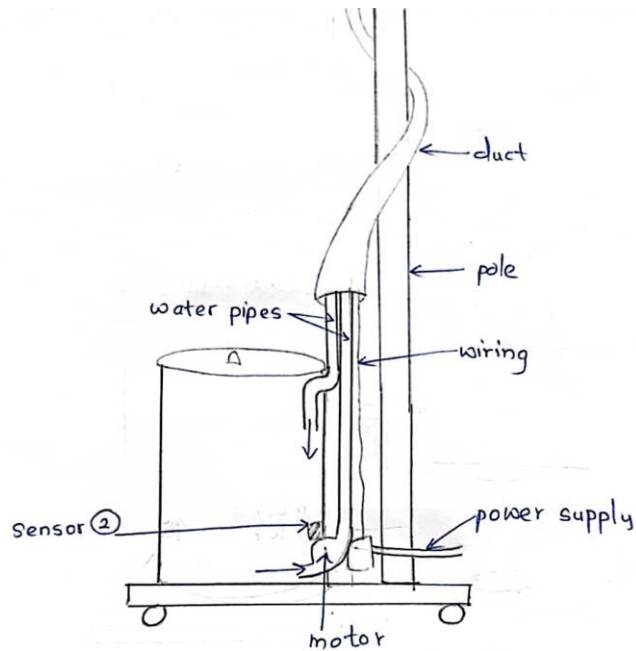
Currently the most popular cooling systems are fans and ac machines. Fans may be inexpensive, but they don't cool down the surrounding environment, just move away the hot air. A/C machines, despite bringing down temperature quite efficiently, are very expensive and hard to afford.

Therefore, there is a gap in the industry to create a sustainable cooling machine that is affordable, efficient, energy saving and environmentally friendly.

2.Introduction the Concept







This is a cooling unit that functions using a normal fan with some added features to improve its cooling ability to make it work like a water cooling fan. The most important feature of the design concept is that this cools water without using electricity and uses that water to bring the room temperature down. It has two main parts connected by a rod and a duct.

3. Working Techniques

3.1 The fan unit

This part consists of a fan and a thin film of cloth to increase the humidity of the air. There is a water container above the cloth and it leaks water to the cloth. Thus humidity of the surrounding air increases. The water is cooled down, so the temperature of the air decreases.

This air flows through the fan unit to the room, thus making the room cooler. There are two ways that reduces the temperature of the room,

1. By moving the hot air and replacing it with new fresh air.
2. By adding humidified and cooler air to the room.

There is a rotating front to the fan. This allows the fan to flow the air everywhere in the room. This unit is connected to a pole with two clamps so that the fan unit can be adjusted to a preferred height. The thin film of cloth is in a zig-zag shape by using nylon threads.

3.2 The cooler unit

This part mainly consists of a clay pot. Storing water in a clay pot drops the temperature of the water thus naturally cooling the water flow for the machine. The clay pot is on a metal stand that has a metal pole. The cooling system can be connected to the power through this unit. There are wheels connected to the metal stand so that the device can be moved easily.

3.3 Both units work together.

The fan unit is connected to a pole that is connected to the stand in the cooling unit. High weight of the clay pot and water inside keeps the whole machine balanced. There is a duct connecting the two parts carrying water and power. Inside the duct are two water pipes and wires.

One pipe carries water up to the water bucket above the thin film. Motor in the clay pot pumps up water. There is another bucket under the cloth that collects leftover water that is brought to the clay pot by another water pipe.

The length of the duct is equal to the height of the pole (1m). It is circled around the pole so that the excess length when the fan unit is in the middle of the pole is kept neatly.

3.3 Power supply to the device

This unit can be connected to the domestic power via a wall outlet. The fan can be directly connected to line power. The motor, sensors and the LED indicator works with an Arduino board. Power needed for the fan unit is taken through the duct.

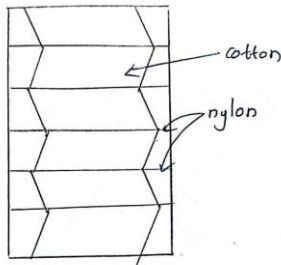
4. Making the Conceptual Design

4.1 Making the stand and the pole.

Aluminum is used to create the metal stand (15cmx15cm) and the pole(1m) because it is strong, inexpensive and light. The clamps are also made from the same material. Wheels are also connected to the metal stand. A store-bought clay pot can be used to store water and we should carefully drill it to make needed holes for water pipes and wiring.

4.2 Making thin film of cloth.

Cotton fabric and nylon thread are used. The frame would be 10"x10" to fit with the fan.



4.3 Making the fan unit.

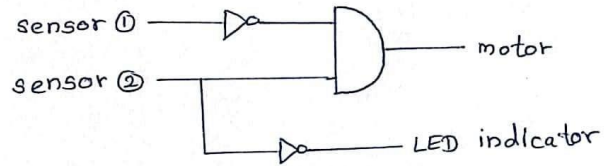
The fan is a normal one that can be bought in a store. The box can be made using hard plastic and attached together. We can use an old ice cream tub for the water bucket. An excess of cloth is left on the water bucket to absorb water. Sensor 1 is connected right above that fabric. A switch and a LED indicator are also attached on the top of the box.

4.4 Attach sensors and the motor

Sensor 1 - checks water level in the top water bucket

Sensor 2 - checks water level in the below clay pot

sensor ① and ②
 1 - there's water
 0 - there's no water



This can be set up using an Arduino board.

5. Efficiency

5.1 Energy

The fan used is a Optimus F-1022 10" electric fan with a wattage of 25W while a normal water cooler consumes 150-300W.

5.2 Cost

Item	Cost (Rs)
Fan	7500
Clay water pot	1000
Cotton fabric	100
Motor	260
Sensors	2x 155
Aluminum	500

Wheels (4)	750
Rubber tubes(4m)	500

Maximum price = Approximately Rs 15,000

While normal water-cooling machine is Rs. 50,000 - 100,000

6. Maintenance and safety in design

Maintenance of this device is very easy because its design is easy to understand and therefore easy to take apart, make corrections and put together again. The parts that will get dirty easily are fabrics and two water containers. These parts are also connected in a way that they can be removed without breaking the whole device, cleaned, and put together again.

The clay pot is at ground level, so nobody will be greatly harmed if it falls down. All the devices in the top fan unit are made of light materials so there won't be an issue there either. The pole is made from aluminum which is light and strong. The device will be in equilibrium because the cooling unit is very heavy with a water filled clay pot so it would not fall over. There won't be any accidents when moving the device because there are wheels attached to make moving easier and safer.

The clay pot is covered by a cover so that it will not be a mosquito breeding ground.

7. Sustainability

This device uses a clay pot to naturally take down water's temperature. It can be also created using components found in day-to-day life and some reusable components.

DESIGN OF A LOW-COST COOLING SYSTEM AS A
SOLUTION FOR THE HIGH COST OF COOLING DEVICES

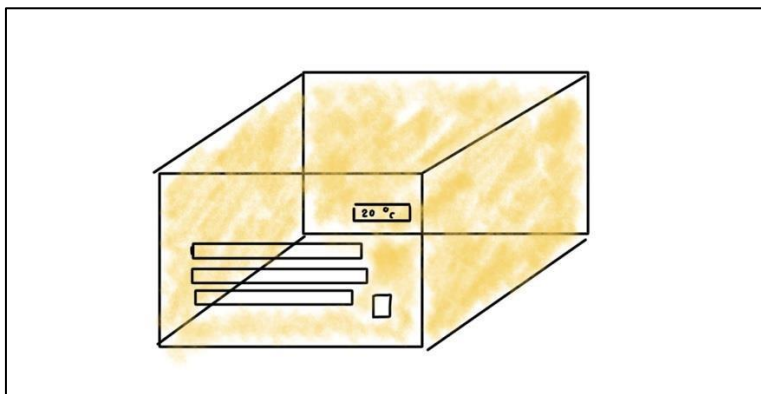
DESIGNED BY,
R.K.P. RANAWEERA
210525C

1. Summary of the problem

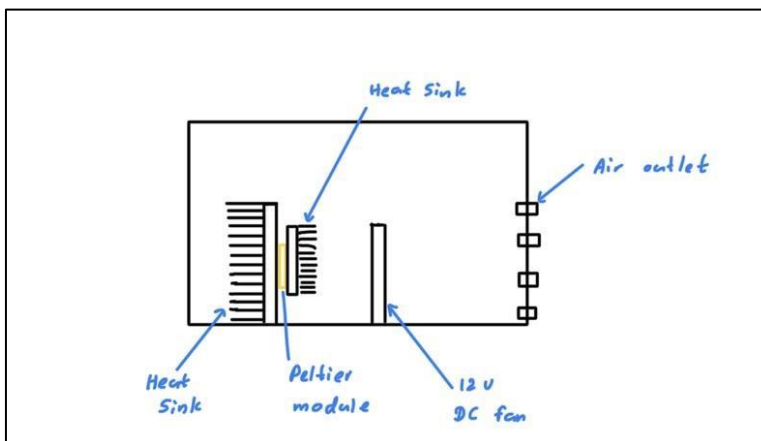
The high cost of traditional air conditioning (AC) systems is a major challenge, limiting their accessibility due to installation, maintenance, and energy expenses. The task is to design a cost-effective alternative cooling system that provides comparable performance and comfort while reducing financial burdens and environmental impact.

2. Introduction to the concept

Here air from the room is taken through the holes and then cooled using the equipment in the design



Outside view



Side view

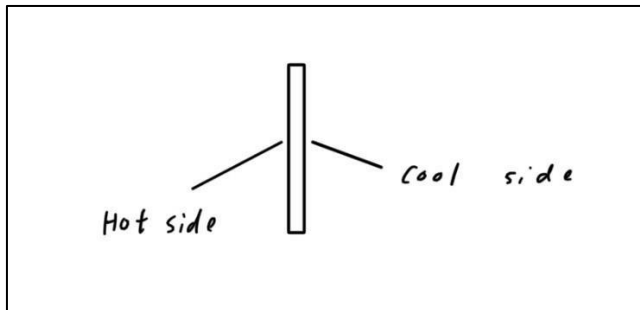
Working techniques

3.1 Peltier module

A Peltier module, also known as a thermoelectric cooler (TEC), is a solid-state device that utilizes the Peltier effect to transfer heat from one side of the module to the other when an electric current is applied. It consists of two ceramic plates with a series of P-type and N-type semiconductor elements sandwiched between them.

The Peltier effect is based on the principle that when an electric current passes through the junction of two dissimilar conductors, heat is absorbed or released, depending on the direction of the current. The Peltier module operates by utilizing this effect.

The Peltier module has two sides: the hot side (also called the high-temperature side) and the cool side (also called the low-temperature side). When the current flows through the module, heat is absorbed from the cool side and transferred to the hot side, resulting in cooling on the cool side and heating on the hot side.



Peltier module

3.2 Heat sink

Cool side heat sink - The heat sink on the cool side of the Peltier module absorbs heat from object or area, spreads it over a larger surface area, and dissipates it into the surrounding air to cool the module and the environment.

Hot side heat sink - The heat sink on the hot side of the Peltier module absorbs heat, dissipates it through an extended surface area, and reduces thermal resistance to prevent overheating.

Making the conceptual design

4.1 Making the outer shell

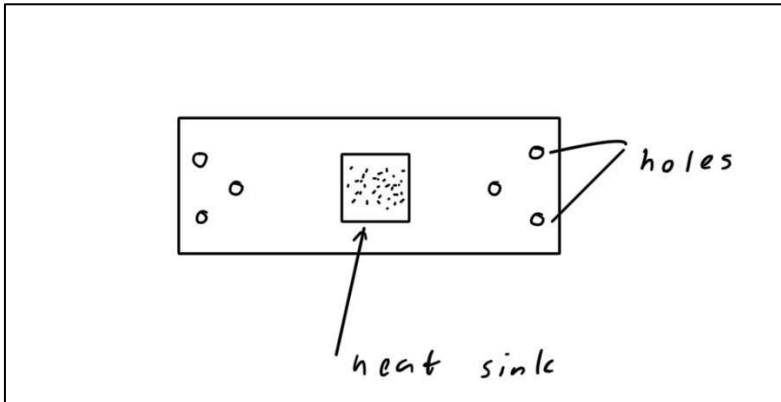
Here we can build the outer shell of the cooler using hard boards. In the front their will be window type spaces that can be used to let air out according to preference. And also, there will be holes to set the temperature indicator and the power on/off button.

4.2 Setting up the equipment

Then inside the box we attach the 12V DC fan to the box.



Then to the back frame of the box we attach a larger heat sink to the outside and then the hot side of the Peltier module is attached to it using thermal paste. And then we attach the smaller heat sink to the cooler side. And then holes should be created in the corners of the back frame.



Finally, we connect a heat insulated pipe to the heat sink to transfer the hot air out.

4.3 Setting up the power supply

Then we have to connect all the equipment using an adapter.

Efficiency

5.1 Energy

The energy source used here is the electricity. We must connect the adapter to the electricity.

5.2 Cost

Hard Board – Rs. 125

Peltier module – Rs. 1200

Heat sink – Rs. 3200

Temperature sensor – Rs. 450

Heat insulating pipe – Rs. 2300

On/off switch – Rs. 100

Total cost of the product will vary around 7300 to 7800.

Rathnamalala T.N.S

DESIGNED BY,
Name:Rathnamalala T.N.S
Index:210533A
Group: Tech Pioneers

1.Summary of the Problem

The current challenge is to develop a cooling mechanism that satisfies three key requirements: affordability, energy efficiency, and effectiveness in cooling indoor areas. Many of the cooling systems available in the market can be costly to purchase and maintain, making them inaccessible to certain consumers. Furthermore, there is a growing need for sustainable cooling alternatives that reduce energy consumption and environmental impact. Consequently, there is a demand for a cooling device that strikes a balance between cost-effectiveness, efficiency, and performance, while also being eco-friendly.

2.Introduction the Concept

In our daily life, energy is wasted in various ways. I propose a smart cooling system named "Smart Green Cooler" using modern Ai technology to reduce the unnecessary energy losses caused by cooling systems . This system can be developed for any cooling system such as cooling fans, AC, and here I consider developing an air conditioner for convenience. This system detects whether there are people in the room and automatically turns off the air conditioner if there are no people, and the temperature and humidity of the room are detected and the temperature given by the air conditioner is automatically changed accordingly.

3. Working Techniques

"Smart Green Cooler" is mainly performed by making recognize whether there are people in the room or not, and if there is no one in the room, the cooling systems (AC, Cooling Fan) are turned off and the conditions of the cooling systems are adjusted according to the temperature and humidity conditions of the room automatically.As mentioned earlier This device has four main 3 Part.

3.1 Occupancy Sensors .

Occupancy sensor detects whether there are people in the room and automatically turns off the air conditioner if there are no people in the room. This usually uses one or more technologies such as infrared, ultrasonic or microwave to detect the presence of people.

3.2 Temperature and humidity sensors.

A temperature and humidity sensor is a sensor system that combines a temperature sensor and a humidity sensor. Temperature sensors work by detecting changes in the temperature of the surrounding

environment. They usually use a thermistor or thermocouple to measure the temperature and then transmit the data to a control system. On the other hand, humidity sensors measure the amount of moisture or water vapor in the air. They can use different technologies such as capacitive, resistive or thermal to measure humidity and send the data to a control system.

3.3 Air Condition

The room is cooled according to the data provided by the occupancy sensor and the temperature and humidity sensor.

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4.2 Making the Occupancy Sensors.

choose a microcontroller: First, you should select a microcontroller that you will use to read data from the sensor and perform any necessary computations. You can choose from a range of popular options such as Arduino, Raspberry Pi, or ESP8266.

Select a motion sensor: Next, you should choose a suitable motion sensor that can detect movement and occupancy. Two of the most commonly used sensors are the PIR (Passive Infrared) and ultrasonic sensors, which can be easily connected to a microcontroller.

Connect the sensor to the microcontroller: Follow the manufacturer's instructions to connect the motion sensor to the microcontroller. Typically, this involves connecting the power, ground, and data pins of the sensor to the corresponding pins on the microcontroller.

Write the code: You will need to write the code to read data from the sensor and detect occupancy. The code can be as simple as detecting a change in the motion sensor data or can involve more complex algorithms to differentiate between people and pets.

Test and calibrate the sensor: Once you have written the code, it's important to test the sensor to ensure that it is functioning properly. You may need to calibrate the sensor to ensure accurate readings.

Enclose the sensor: Finally, you can enclose the sensor in a protective case and attach it to a wall or other surface as needed

4.3 setup occupy sensor and humidity and temperatur sensor to Ac

To set up an occupancy sensor and temperature/humidity sensor to an AC unit, you will need to follow these steps:

1. Connect the sensors to a microcontroller: You will need to connect the occupancy sensor and temperature/humidity sensor to a microcontroller like an Arduino or Raspberry Pi. Follow the manufacturer's instructions to properly connect the sensors to the microcontroller.
2. Write the code: You will need to write the code that will read the data from the sensors and send commands to the AC unit based on the readings. For example, if the occupancy sensor detects someone in the room, the code can turn on the AC unit. If the temperature/humidity sensor detects that the temperature or humidity is too high, the code can turn on the AC unit and set it to a lower temperature.
3. Connect the microcontroller to the AC unit: You will need to connect the microcontroller to the AC unit to send commands. This can be done using an infrared (IR) transmitter or using a smart AC unit that can be controlled through WiFi.
4. Test and calibrate the system: Once you have written the code and connected the microcontroller to the AC unit, it's important to test the system to ensure that it is functioning properly. You may need to calibrate the sensors to ensure accurate readings.
5. Enclose the system: Finally, you can enclose the microcontroller, sensors, and any other components in a protective case and attach it to a wall or other surface as needed.

5. Efficiency

5.1 Energy

An occupancy and temperature/humidity sensor system can be more energy-efficient than traditional AC units because it can automatically adjust the AC settings based on the current conditions in the room. By doing so, it can avoid overcooling or overheating a space, which can waste energy and increase your energy bill.

Here are some ways that an occupancy and temperature/humidity sensor system can help improve energy efficiency:

1. Avoiding unnecessary cooling or heating: With traditional AC units, it's common for people to forget to adjust the thermostat when they leave a room, which can lead to unnecessary cooling or heating. With an occupancy sensor, the system can detect when a room is unoccupied and adjust the AC settings accordingly, which can save energy.
2. Personalized comfort: With an occupancy and temperature/humidity sensor system, the AC settings can be adjusted based on the number of people in the room and their temperature preferences. This can help ensure that the space is only cooled or heated to the level necessary to maintain a comfortable environment, which can save energy.
3. Automated control: With an occupancy and temperature/humidity sensor system, the AC settings can be automatically adjusted based on the current conditions in the room. This eliminates the need for manual adjustments, which can help prevent overcooling or overheating a space.

Overall, an occupancy and temperature/humidity sensor system can help improve energy efficiency by reducing unnecessary cooling or heating, personalizing comfort, and offering automated control. By doing so, it can help you save energy and reduce your energy bill compared to traditional AC units.

5.2 Cost

The cost efficiency of an occupancy and temperature/humidity sensor system compared to traditional AC units will depend on several factors, including the initial cost of the system, the cost of installation and maintenance, and the energy savings that can be achieved.

While an occupancy and temperature/humidity sensor system may have a higher initial cost compared to traditional AC units, it can potentially provide cost savings over the long term due to the energy savings it can achieve. By avoiding unnecessary cooling or heating and personalizing comfort, an occupancy and temperature/humidity sensor system can help reduce energy consumption and lower energy bills.

Moreover, an occupancy and temperature/humidity sensor system can potentially extend the lifespan of the AC units by preventing overuse and avoiding excessive wear and tear. This can lead to lower maintenance costs over the long term.

Therefore, it's difficult to provide a specific cost efficiency comparison between an occupancy and temperature/humidity sensor system and traditional AC units, as the cost savings will depend on the specific circumstances and usage of the system. However, by reducing energy consumption and avoiding unnecessary wear and tear, an occupancy and temperature/humidity sensor system can potentially provide cost savings over the long term.

Below is an approximate initial cost to install this system.

Part	Cost(LKR)
Occupancy sensor	5000
Temperature and humidity sensor	2500
Controller	7000
Total	14500

6. Maintenance and safety in design

When designing and using an occupancy sensor and temperature/humidity sensor system, it's important to consider maintenance and safety. Here are some factors to keep in mind: Maintenance:

1. Regularly check the sensors for proper functioning and accuracy. Dust and debris can accumulate on the sensors, so it's important to clean them periodically.
2. Test the system regularly to ensure that it is working properly.
3. Update the software as needed to fix bugs or improve performance.
4. Replace any worn-out or damaged components promptly to avoid system failure.

Safety:

7. 1. When working with electricity, always follow safety guidelines and take necessary precautions.
2. Ensure that the system is properly grounded to avoid electrical shock.
3. Avoid placing the system in areas where it may be exposed to water or other liquids.
4. Make sure the system is not obstructing any walkways or posing any tripping hazards.

Ease of Design

Compared to traditional AC units, an occupancy and temperature/humidity sensor system can offer several benefits in terms of ease of use. Here are some reasons why:

1. Automated control: An occupancy and temperature/humidity sensor system can automatically adjust the AC settings based on the current conditions in the room. This eliminates the need for manual adjustments, making it more convenient and easier to use.
2. Energy efficiency: By automatically adjusting the AC settings based on the current conditions, an occupancy and temperature/humidity sensor system can be more energyefficient than traditional AC units. This can help reduce energy costs and lower your carbon footprint.
3. Personalized comfort: An occupancy and temperature/humidity sensor system can provide personalized comfort by adjusting the AC settings based on the number of people in the room and their temperature preferences. This can help ensure that everyone in the room is comfortable, regardless of individual preferences.
4. Remote control: Many occupancy and temperature/humidity sensor systems can be controlled remotely using a mobile app or other device. This makes it easy to adjust the AC settings from anywhere, which can be especially convenient when you are away from home.

Overall, an occupancy and temperature/humidity sensor system can be easier to use than traditional AC units because it offers automated control, energy efficiency, personalized comfort, and remote control capabilities. However, it's important to note that the ease of use will depend on the specific design and implementation of the system.

8. Sustainability

The sustainability of an occupancy and temperature/humidity sensor system will depend on several factors, including the materials and components used in the system, the energy efficiency it provides, and the impact it has on the environment during production, installation, and disposal.

Here are some ways in which an occupancy and temperature/humidity sensor system can contribute to sustainability:

- Energy efficiency: As mentioned earlier, an occupancy and temperature/humidity sensor system can help improve energy efficiency and reduce energy consumption. By doing so, it can help reduce greenhouse gas emissions and mitigate the impact of climate change.

- Improved indoor air quality: An occupancy and temperature/humidity sensor system can help ensure that the air conditioning system is not overused, which can help maintain better indoor air quality by avoiding the buildup of pollutants and allergens.
- Reduced waste: By helping to extend the lifespan of the AC units, an occupancy and temperature/humidity sensor system can help reduce the amount of waste generated by the replacement of old and worn-out units.
- Sustainable materials and components: Using sustainable materials and components in the system, such as energy-efficient sensors and controllers, can further contribute to sustainability by reducing the environmental impact of production and disposal.

Overall, an occupancy and temperature/humidity sensor system can contribute to sustainability by improving energy efficiency, indoor air quality, and reducing waste.

THE NEED FOR COOLING SYSTEMS DUE TO HIGH HEAT,
BUT THESE SYSTEMS BEING TOO EXPENSIVE

Sandaru H.W.P

Index No:210564T

1. Problem

the need for cooling systems due to high heat, but these systems being too expensive.

2. purpose of the project

design a low-cost cooling system for high heat.

introduction to concept

After conducting a problem analysis, our team discovered that high temperatures in rooms are a common issue and traditional cooling systems can be prohibitively expensive. Therefore, our main objective was to find an affordable cooling solution. As a result, I proposed the idea of an evaporative air cooler called “air chiller”.

Description about the cooler

- 12 V DC Fan (2)

12V fan is an electric fan that operates on a 12-volt direct current power supply. It typically has small blades that rotate at high speed to move air and provide ventilation or cooling. So, the cooled vapor is ventilated by these fans.



- Mist maker

A mist maker, also known as a fogger or humidifier, is a device that creates a fine mist by forcing water through a small orifice using high-pressure ultrasonic vibrations. In an evaporative cooling system, the purpose of a mist maker is to increase the humidity level of the air to enhance the cooling effect.

“Air chiller” cooling systems work by using the natural process of water evaporation to cool down the air. As water evaporates, it absorbs heat from the surrounding air, thereby reducing the air temperature. In a typical evaporative cooling system, a fan draws hot, dry air through a water-soaked pad or filter. As the air passes through the pad, the water evaporates and cools the air, which is then blown back into the room.

By adding a mist maker to the system, the humidity level of the air can be increased, which improves the effectiveness of the cooling process. The fine mist created by the mist maker also helps to distribute the water more evenly across the cooling pad, which can result in more efficient cooling. Additionally, the mist maker can help to purify the air by removing dust and other particles, which can improve indoor air quality.

Overall, the purpose of a mist maker in “Air chiller” cooling system is to enhance the cooling effect by increasing humidity levels and improving the distribution of water across the cooling pad.



- Box made of plastic.

This box contains water and ice. To maintain the low temperature of the water, it is recommended to use a plastic box with low heat insulation. The plastic box will create a barrier that will prevent heat from entering the box and melting the ice, thus keeping the water at a low temperature. the top of the plastic box includes two holes for air out and two holes for air in and one hole for put water(/ice).



- L shape Plastic tubes

Two L shape plastic tubes were attached to the top of the box. through these tubes cooled air will be removed to the outside



- 12 V DC adaptor

The purpose of a 12V DC adaptor is to convert the AC (alternating current) power from a wall outlet into the DC (direct current) power that the fan requires to operate. The adaptor typically provides a stable and consistent 12V DC power supply, which is necessary for the fan to function properly and efficiently.

Without the adaptor, the fan would not be able to operate as it requires a 12V DC power supply to function. Using a different power supply with an incorrect voltage or polarity can cause damage to the fan or result in the fan not working properly. Therefore, it is important to use a 12V DC adaptor that is specifically designed for the fan to ensure its safe and proper operation.



- Switch

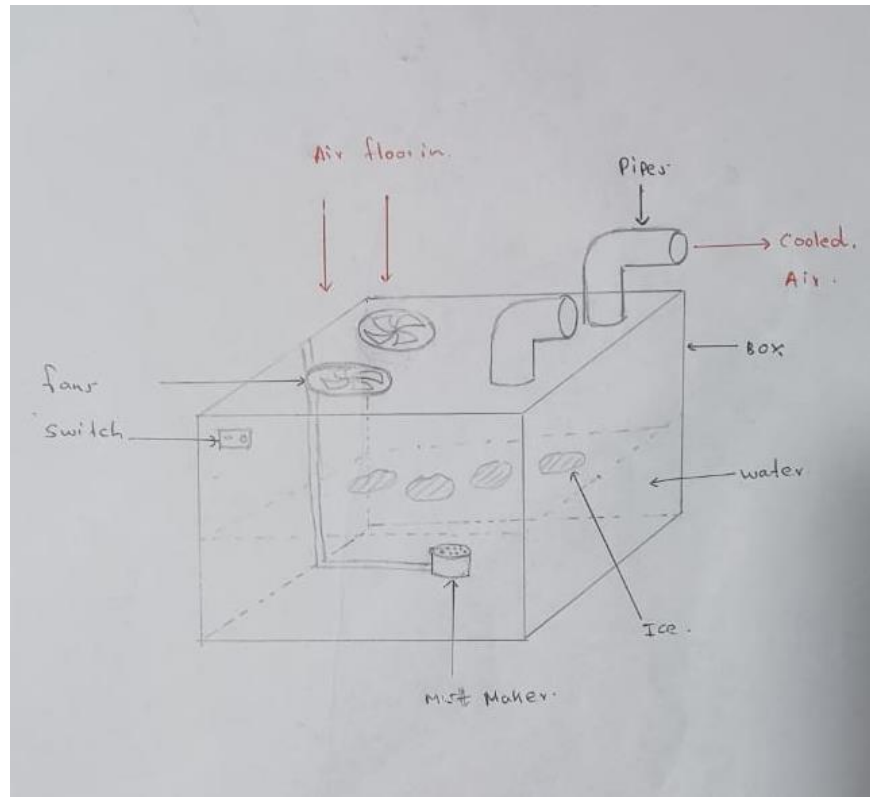
The purpose of a switch in this system is to control the operation of the system by turning it on or off based on specific conditions or parameters. switch may be used to turn on or off the fans and the mist maker of the system.



- Wires

Wires are used to connect the electric components.





Using method

- Prepare the cooler: Make sure the water tank is filled with clean water(/ice)
- Place the cooler: Place the evaporative cooler in a well-ventilated area, ideally near an open window or door to allow fresh air to circulate.
- Turn on the cooler: Plug in the cooler and turn it on by on/off switch.
- Enjoy the cool air: Once the cooler is running, it will begin to circulate cool, moist air. Make sure to keep the room well-ventilated, either by opening windows or doors, or by using a vent or exhaust fan, to prevent the air from becoming too humid.

- Maintain the cooler: Clean the cooler regularly to prevent the buildup of mineral deposits or bacteria. Empty and refill the water tank as needed with water(/ice).

Problems of the design

- Take sometimes to work properly.
- Water(/ice) needs to be refilled.

Solutions

- Replace fan with a power full one.
- Box can be connected to a water source and set a device to fill the box occasionally with the low amount of water.

Reference

- J. Wong, "What is mist maker," Beijing Ultrasonic, 25-Apr-2023. [Online]. Available: <https://www.bjultrasonic.com/what-is-mist-maker/>. [Accessed: 13-May2023].
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- Wikipedia contributors, "Thermal insulation," Wikipedia, The Free Encyclopedia, 09-Apr-2023. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Thermal_insulation&oldid=1149012_363.

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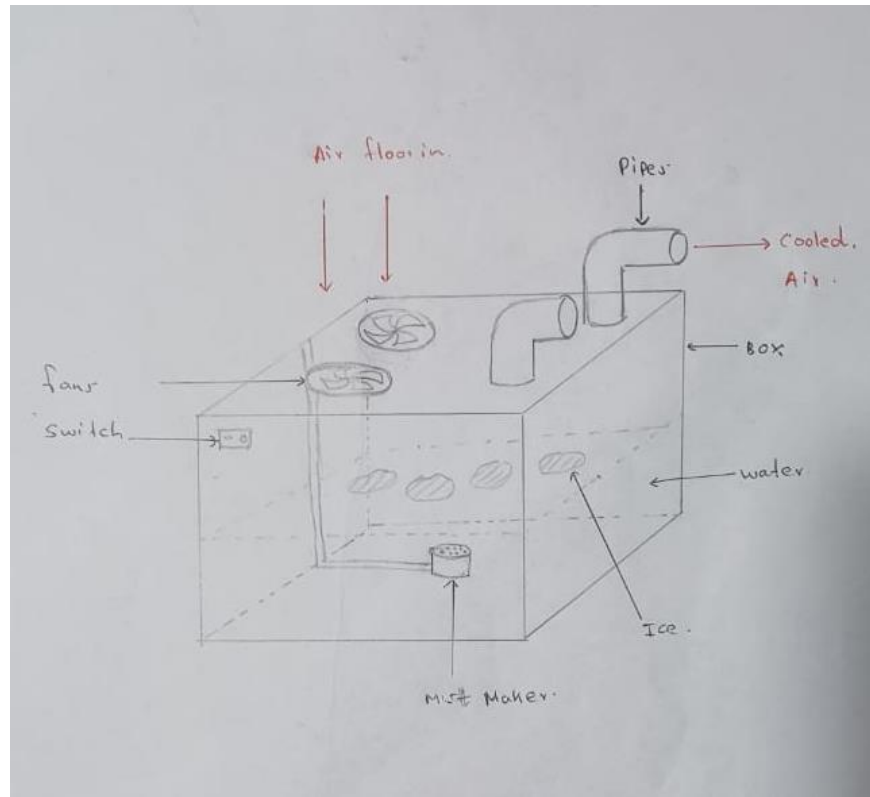


- Switch
The purpose of a switch in this system is to control the operation of the system by turning it on or off based on specific conditions or parameters. switch may be used to turn on or off the fans and the mist maker of the system.



- Wires
Wires are used to connect the electric components.





Using method

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- Place the cooler: Place the evaporative cooler in a well-ventilated area, ideally near an open window or door to allow fresh air to circulate.
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- Wikipedia contributors, "Thermal insulation," Wikipedia, The Free Encyclopedia, 09-Apr-2023. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Thermal_insulation&oldid=1149012_363.

Appendix II: Table of Contributions from Each Member

Index No	Name	Work done
210410U	NAVINNA R.R.	Work for prototype, presentation, report
210447M	PATHIRAGE S.S.K.	Work for prototype, presentation, report
210458X	PERERA G.D.D.C.	Work for prototype, presentation, report
210496K	PREMARATHNE O.D.	Work for prototype, presentation, report
210494D	PREMAKUMARA H.P.S.	Work for prototype, presentation, report
210509G	RAJAPAKSHA W.R.A.K.H.	Work for prototype, presentation, report
210525C	RANAWEERA R.K.P.	Work for prototype, presentation, report
210533A	RATHNAMALALA T.N.S.	Work for prototype, presentation, report
210564T	SANDARU H.W.P.	Work for prototype, presentation, report

stakeholder and corresponding team member name and index number

Index No	Name	Stakeholder
210410U	NAVINNA R.R.	repair shops
210447M	PATHIRAGE S.S.K.	transportation services
210458X	PERERA G.D.D.C.	religious places
210496K	PREMARATHNE O.D.	university/schools
210494D	PREMAKUMARA H.P.S.	government
210509G	RAJAPAKSHA W.R.A.K.H.	residents
210525C	RANAWEERA R.K.P.	manufacturing/distribution companies
210533A	RATHNAMALALA T.N.S.	other businesses
210564T	SANDARU H.W.P.	restaurants and hotels

8.3.Appendix III: Meeting Minutes

Team Name: Tech Pioneers

Date and Time: 13th March 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: RATHNAMALALA T.N.S.

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G

2. Goals

1. Introducing the team members
2. Selecting a leader
3. Discussing and agreeing to team rules

3. Minutes

- a. PREMARATHNE O.D. nominated PREMAKUMARA H.P.S. as the leader.
- b. RANAWEERA R.K.P. confirmed that.
- c. Each member was tasked with finding problems due on 17th March 2023.

Team Name: Tech Pioneers

Date and Time: 20th March 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: RATHNAMALALA T.N.S.

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	PATHIRAGE S.S.K.	210447M
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G

2. Agenda Topics

1. Apologies
2. Project brief

3. Goals

Get ideas from each team member and discuss with each other to find a problem so that a solution can be found.

4. Minutes

a. Apologies

PATHIRAGE S.S.K apologized for not attending the previous meeting on 13th

March 2023.

b. Project brief

Each member shared their ideas and everybody was asked to think and research more about these ideas.

High electricity in the present situation and high cost for cooling devices are the major problems after discussing.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Presented the problem of higher electricity bills at the present time. Gave an idea about a fan that turns on automatically when the temperature is too high and low cost.	Find more details about discussed problems and ideas	27 th March 2023
PATHIRAGE S.S.K	Gave an idea about making the student chairs used at the university easier to use.	Find more details about discussed problems and ideas	27 th March 2023
PERERA G.D.D.C.		Find more details about discussed problems and ideas	27 th March 2023

PREMAKUMARA H.P.S.	Gave an idea about a mouse that can operate wired, wireless and Bluetooth. Gave an idea about making color lights turn red as soon as the pedestrians crossed the road. Gave an idea about a thinner laptop cooling pad. Gave an idea about generating electricity from a stress ball. Gave an idea about preserving food, especially vegetables without a refrigerator. Give an idea about Currently, most of the electricity is spent on cooling and the devices used for this are very expensive	Find more details about discussed problems and ideas	27 th March 2023
PREMARATHNE O.D.	Presented the problem of poor waste management of the university.	Find more details about discussed problems and ideas	27 th March 2023
RAJAPAKSHA W.R.A.K.H.	Gave an idea about working on the bed more comfortably.	Find more details about discussed problems and ideas	27 th March 2023
RANAWEERA R.K.P.	Gave an idea about a mug that reheats by itself.	Find more details about discussed problems and ideas	27 th March 2023

RATHNAMALALA T.N.S.	Gave an idea about developing a new harmless material for recharge card coatings.	Find more details about discussed problems and ideas	27 th March 2023
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Team Name: Tech Pioneers

Date and Time: 27th March 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: RATHNAMALALA T.N.S.

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G
PATHIRAGE S.S.K.	210447M		

2. Agenda Topics

1. Apologies
2. Approval of the previous meeting
3. Project brief

3. Goals

1. Selecting a title for the project
2. Make a gantt chart

4. Minutes

1. Apologies

SANDARU H.W.P apologized for not attending the previous meeting on 20th March 2023.

- b. Project brief

End of the session we decide the our main problem as high cost for cooling device

Every member agreed to the idea that the title of the project would be to build a cooling system for a comfortable price.

A gantt chart for the project was created.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023
PATHIRAGE S.S.K	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023
PERERA G.D.D.C.	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023
PREMAKUMARA H.P.S.	Created the gantt chart.	Finding relevant information about the project.	3 rd April 2023
PREMARATHNE O.D.	Created the gantt chart.	Finding relevant information about the project.	3 rd April 2023

RAJAPAKSHA W.R.A.K.H.	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023
RANAWEERA R.K.P.	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023
RATHNAMALALA T.N.S.	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023
SANDARU H.W.P.	Gave ideas on allocating time in the gantt chart.	Finding relevant information about the project.	3 rd April 2023

Team Name: Tech Pioneers

Date and Time: 2nd April 2023. 8.00pm – 9.30pm

Mode: Zoom Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: RATHNAMALALA T.N.S.

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G
PATHIRAGE S.S.K.	210447M		

2. Agenda Topics

1. Approval of the previous meeting
2. Project brief

3. Goals

- a. Finalizing the gantt chart

4. Minutes

- a. Project brief

The sketch of the gantt chart that was created during the last meeting was discussed again and finalized.

Team Name: Tech Pioneers

Date and Time: 3rd April 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: RATHNAMALALA T.N.S.

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G
PATHIRAGE S.S.K.	210447M		

2. Agenda Topics

1. Approval of the previous meeting
2. Project brief

3. Goals

1. Deciding what information each member should be collected

4. Minutes

1. Project brief

What kind of information should be collected was decided and each member picked a subtopic to gather information.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Decided that information about how long an alternative cooling system will work should be gathered.	To find information about how long an alternative cooling system	10 th April 2023
PATHIRAGE S.S.K	Decided that how the current cooling systems affect the electricity bill should be studied.	To find information on how the current cooling systems affect the electricity bill	10 th April 2023
PERERA G.D.D.C.	Decided information on how cooling systems are maintained should be gathered.	To find information on how cooling systems are maintained	10 th April 2023
PREMAKUMARA H.P.S.	Decided that information about different cooling systems should be gathered.	To find information on different cooling systems	10 th April 2023
PREMARATHNE O.D.	Decided that more information on how ac machines work should be found.	To go to a repairing shop of ac machines and ask how these work	10 th April 2023

RAJAPAKSHA W.R.A.K.H.	Decided that information on different climates in different regions should be gathered.	To find information on different climates in different regions	10 th April 2023
RANAWEERA R.K.P.	Decided that information on current alternative cooling systems should be found.	To find information on current alternative cooling systems	10 th April 2023
RATHNAMALALA T.N.S.	Decided that we should find out whether there are cooling systems without using electricity.	To find information on whether there are cooling systems without using electricity	10 th April 2023
SANDARU H.W.P.	Decided that prices of current cooling systems should be found.	To find information on prices of current cooling systems	10 th April 2023

Team Name: Tech Pioneers

Date and Time: 19th April 2023. 9.00pm – 10.00pm

Mode: Zoom Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMAKUMARA H.P.S

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G
PATHIRAGE S.S.K.	210447M		

2. Agenda Topics

1. Apologies
2. Approval of the previous meeting
3. Project brief

3. Goals

1. Prepare the stakeholder analysis

4. Minutes

1. Apologies

PATHIRAGE S.S.K. apologized for being late for the meeting.

- b. Project brief

We collected and discussed gathered information. Some more information was to be found. A stakeholder list was prepared and distributed among the group to analyze them.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Preparing the stakeholder list.	To analyze repair shops of cooling systems as stakeholders	20 th April 2023
PATHIRAGE S.S.K		To analyze transportation services as stakeholders	20 th April 2023
PERERA G.D.D.C.	Preparing the stakeholder list.	To analyze religious places as stakeholders	20 th April 2023
PREMAKUMARA H.P.S.	Preparing the stakeholder list. Gave previously gathered relevant information Decided that we should find more information on what are the questions asked before installing an AC system in a building.	To analyze government as stakeholders	20 th April 2023

PREMARATHNE O.D.	Preparing the stakeholder list.	To analyze university/schools as stakeholders To visit an AC shop	20 th April 2023
RAJAPAKSHA W.R.A.K.H.	Preparing the stakeholder list. Gave previously gathered relevant information	To analyze residents as stakeholders	20 th April 2023
RANAWEERA R.K.P.	Preparing the stakeholder list.	To analyze manufacturing/distribution companies of cooling systems as stakeholders	20 th April 2023
RATHNAMALALA T.N.S.	Preparing the stakeholder list.	To analyze other businesses such as office, factories as stakeholders	20 th April 2023
SANDARU H.W.P.	Preparing the stakeholder list.	To analyze restaurants and hotels as stakeholders	20 th April 2023

Team Name: Tech Pioneers

Date and Time: 23th April 2023. 10.00pm – 10.30pm

Mode: Zoom Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMAKUMARA H.P.S

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
RAJAPAKSHA W.R.A.K.H.	210509G	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K		

2. Agenda Topics

1. Approval of the previous meeting
2. Project brief

3. Goals

1. Finalize the stakeholder analysis
2. Solve some problems regarding the project

4. Minutes

1. Project Brief

PREMAKUMARA H.P.S asked some questions that had been raised during the previous meeting from Ms. Dushani and resolved them.

Team Name: Tech Pioneers

Date and Time: 24th April 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMAKUMARA H.P.S

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	PATHIRAGE S.S.K.	210447M
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G

2. Agenda Topics

1. Approval of the previous meeting
2. Project brief

3. Goals

1. Preparing the weighted decision matrix

4. Minutes

1. Project Brief

Some more changes to the stakeholder analysis were made. The timeline for the project was told to us. We decided to start making the conceptual design before the decided date.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
PATHIRAGE S.S.K	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
PERERA G.D.D.C.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
PREMAKUMARA H.P.S.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
PREMARATHNE O.D.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
RAJAPAKSHA W.R.A.K.H.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
RANAWEERA R.K.P.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023

RATHNAMALALA T.N.S.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023
SANDARU H.W.P.	Preparing the weighted decision matrix	Create a conceptual design	12 th May 2023

Team Name: Tech Pioneers

Date and Time: 8th May 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMAKUMARA H.P.S

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PREMAKUMARA H.P.S	210494D	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G
PATHIRAGE S.S.K.	210447M		

2. Agenda Topics

1. Apologies
2. Approval of the previous meeting
3. Project brief

3. Goals

1. Finalizing the weighted decision matrix.
2. Getting an idea on how to write the conceptual design.
3. Discussing each one's conceptual design

4. Minutes

1. Apologies

RATHNAMALALA T.N.S.. apologized for not attending the previous meeting.

- b. Project brief

There were some loopholes in the weighted decision matrix, so it was finalized. We were informed about the deadlines of each task and the structure of the conceptual design report. Then each of us discussed the conceptual design ideas we have so far.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.		Create a conceptual design	12 th May 2023
PATHIRAGE S.S.K		Create a conceptual design	12 th May 2023
PERERA G.D.D.C.		Create a conceptual design	12 th May 2023
PREMAKUMARA H.P.S.		Create a conceptual design	12 th May 2023
PREMARATHNE O.D.		Create a conceptual design	12 th May 2023
RAJAPAKSHA W.R.A.K.H.		Create a conceptual design	12 th May 2023

RANAWEERA R.K.P.		Create a conceptual design	12 th May 2023
RATHNAMALALA T.N.S.		Create a conceptual design	12 th May 2023
SANDARU H.W.P.		Create a conceptual design	12 th May 2023

Team Name: Tech Pioneers

Date and Time: 14th May 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMARATHNE O.D.

1. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PATHIRAGE S.S.K.	210447M	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G

2. Agenda Topics

1. Approval of the previous meeting
2. Project brief

3. Goals

1. Peer reviewing

4. Minutes

1. Project brief

Project leader PREMAKUMARA H.P.S was absent due to dengue fever.

We reviewed other group members' conceptual designs using the weighted decision matrix.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
PATHIRAGE S.S.K	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
PERERA G.D.D.C.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
PREMARATHNE O.D.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
RAJAPAKSHA W.R.A.K.H.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
RANAWEERA R.K.P.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
RATHNAMALALA T.N.S.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023

SANDARU H.W.P.	Peer reviewing	Complete peer reviewing and fill the database	16 th May 2023
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Team Name: Tech Pioneers

Date and Time: 21th May 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMARATHNE O.D.

5. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PATHIRAGE S.S.K.	210447M	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G

6. Agenda Topics

1. Approval of the previous meeting
2. Project brief

7. Goals

1. Choosing a final design
2. Alternating and developing the chosen design with ideas from the group members.
3. Giving each team member a task for making the prototype and writing the report.

8. Minutes

1. Project brief

Project leader PREMAKUMARA H.P.S was absent due to dengue fever.

The conceptual design of PREMARATHNE O.D. was chosen as the design idea. NAVINNA R.R. proposed to combine ideas from her design to the final design. RAJAPAKSHA W.R.A.K.H. approved that idea.

We were given an idea on how to prepare the report and important topics.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Adding ideas to refine the final design.	To bring electronic components needed for the prototype.	28 th May 2023
PATHIRAGE S.S.K	Adding ideas to refine the final design.	To bring other needed components for the prototype.	28 th May 2023
PERERA G.D.D.C.	Adding ideas to refine the final design.	To bring a box to house the prototype.	28 th May 2023
PREMARATHNE O.D.	Adding ideas to refine the final design.	To make the metal frame of the prototype.	28 th May 2023
RAJAPAKSHA W.R.A.K.H.	Adding ideas to refine the final design.	To bring other needed components for the prototype.	28 th May 2023
RANAWEERA R.K.P.	Adding ideas to refine the final design.	To bring other needed components for the prototype.	28 th May 2023

RATHNAMALALA T.N.S.	Adding ideas to refine the final design.	To bring other needed components for the prototype.	28 th May 2023
SANDARU H.W.P.	Adding ideas to refine the final design.	To bring other needed components for the prototype.	28 th May 2023

Team Name: Tech Pioneers

Date and Time: 28th May 2023. 11.15am – 12.15pm

Mode: Physical Meeting

Team Leader: PREMAKUMARA H.P.S

Convener: PREMARATHNE O.D.

9. Attendees

Name	Index No.	Name	Index No.
NAVINNA R.R.	210410U	RANAWEERA R.K.P.	210525C
PERERA G.D.D.C.	210458X	RATHNAMALALA T.N.S.	210533A
PATHIRAGE S.S.K.	210447M	SANDARU H.W.P.	210564T
PREMARATHNE O.D.	210496K	RAJAPAKSHA W.R.A.K.H.	210509G

10. Agenda Topics

1. Approval of the previous meeting
2. Project brief

11. Goals

1. To prepare all the components needed for the prototype.
2. Allocating a date and time to build the prototype and write the report.
3. Give each member a part to write in the report.

12. Minutes

1. Project brief

Project leader PREMAKUMARA H.P.S was absent due to dengue fever.

PREMARATHNE O.D. , NAVINNA R.R , PERERA G.D.D.C. and RAJAPAKSHA W.R.A.K.H. told what items they could gather for the prototype.

A list of needed items was prepared and SANDARU H.W.P. and RANAWEERA R.K.P. volunteered to buy them. A price list was created simultaneously.

30th and 31st were chosen to work on the report and the prototype.

PREMAKUMARA H.P.S was informed of the progress via phone and he decided to bring some items himself.

Summary Table

Name	Work done	Goals	Date to be done
NAVINNA R.R.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023
PATHIRAGE S.S.K	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023
PERERA G.D.D.C.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023
PREMARATHNE O.D.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023
RAJAPAKSHA W.R.A.K.H.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023

RANAWEERA R.K.P.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023
RATHNAMALALA T.N.S.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023
SANDARU H.W.P.	Preparing needed items and planning how to build the prototype.	Prepare the report and build the prototype.	03 rd June 2023